

Cadence: learning through local repair

Persistent patch networks, equilibrium detuning, and the limits of memory

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Draft of September 20, 2026

Abstract

Cadence investigates whether continuing learning can be organized around bounded observer-like patches that retain local state, exchange readings through ports and repair inconsistent predictions. The core combines nonlinear neural activity with persistent relationships learned through local free/nudged equilibrium contrasts. Optional temporal overlap holds previous activity fixed throughout each event; imagined branches are read-only. A readable activity trace is a separate carried boundary. The interface needs no attached episodic store, but useful memory requires evidence beyond numerical convergence. In a four-fact task, supplied noninterfering plasticity permits acquisition, retention and selective correction in all five fresh initializations, while a matched all-plastic control passes none. This uses inherited addresses and fixed shared relationships, with 6.47 times the integration work. On recorded-audio prediction, autoregressive controls outperform both tested neural settings, whose forecast tails become constant. A brief-intention test retains a measurable trace after identical sensory histories but fails to turn it into useful action. Conditional machine-checked results distinguish activation history, readable trace, parameter memory and causal provenance; none establishes autonomous importance or useful specialization. Separate nonlinear diagnostics show that completion can coexist with failed revision and that individually durable contacts can obstruct distributed learning. The architectural hypothesis is a continuing whole-network actor with memory, intention and consequence readback through common patch operations. The tests do not establish a functioning composer, general lifelong memory or biological equivalence.

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1 Introduction

1.1 From cognitive competence to embodied learning

Reasoning and conversational competence warrant taking the case for general-purpose cognitive AI seriously. An advanced Gemini Deep Think system achieved gold-medal-level performance on the 2025 International Mathematical Olympiad [46]; a preregistered five-minute Turing-test study found persona-prompted GPT-4.5 selected as human 73% of the time [35]. Whether such capabilities qualify as AGI depends on the required breadth, performance and autonomy [51]. Neither result measures the ability to learn a changing body or retain skills through prolonged physical interaction.

Here *embodied AGI* denotes robots that can be taught a broad range of unfamiliar tasks, learn from their consequences and retain and transfer useful skills under bounded compute, memory and energy budgets. Robot learning through demonstrations, corrections and autonomous experience has concrete precedents in vision-language-action systems [64]. Cadence’s architectural thesis concerns the organization and cost of that learning: persistent local computation, directly writable experience and adaptable connections should make continuing adaptation more economical at matched competence.

A frozen feed-forward model evaluates a learned map at deployment. Recurrent learners, online gradient methods and transformers with key/value caches provide other forms of carried state; backpropagation does not require learning to stop [69, 78]. The question here is which state a learner should retain and update when its task is an ongoing interaction. A robot can accumulate a model of its own consequences, consult that model before acting and revise it after an unexpected outcome. The cost of this process includes its representation, memory, search and every learning update.

Cadence tests a particular organization of that process: continuing neural state and learned relationships between bounded patches. Earlier applications combined that computation with directly writable sparse records. The revised interface puts recurrent repair and equilibrium detuning at the centre, so that a separate memory mechanism must earn its inclusion through a measured capability. Three questions guide the comparison. Does the system learn useful behaviour from one stream? Does it preserve and revise what it learned? Does reusing state save work at the required accuracy and latency? A dictionary, numerical solver or gradient learner can share several of these advantages.

1.2 OPH foundations and neural influences

Pragma’s physics and consensus research. Cadence’s organizing framework comes from Pragma Research’s Observer-Patch Holography (OPH): bounded, self-reading patches repair disagreements on declared interfaces. The repair and observation-determined normal-form theorems give conditions for consistent global answers independent of repair order and determined by protected observations [58, 59]. Event algebras define the observable distinctions [55]. Section 4.2 gives the neural specialization; its guarantees are independent of OPH’s physical interpretation.

Eliasmith: integrating perception, cognition and action. Eliasmith’s *How to Build a Brain* [20] was an architectural influence on Cadence. Its integrated treatment of perceptual, cognitive and motor models motivates the ambition of a neural system that behaves as a whole. The arm-and-eye demonstration was inspired by examples on Eliasmith’s website.¹

Hopfield: computation by settling. Hopfield’s graded-response networks compute through collective relaxation and associative recall [32]. Cadence uses nonlinear recurrent settling. A sufficiently small incoming-weight bound certifies a unique fixed-drive equilibrium and a residual error bound. Outside that subclass, multiple basins may support history-dependent activity, but the contraction certificate cannot be invoked. Neither regime establishes biological fidelity.

Equilibrium propagation: local credit. Scellier and Bengio derive objective gradients from free and nudged equilibria under stated symmetry and smoothness assumptions [70]. The centred positive/negative

¹See the group’s Semantic Pointer Architecture and Spaun demonstrations.

estimator follows Laborieux et al. [42]. Cadence applies these local contrasts and measures their state-storage cost and learning behaviour; the perturbation phases must also be computed.

Marr and Albus: expansion and memory. Cerebellar theories motivate expanded representations with a plastic readout [2, 47]. Here sparse keys address residual record writes. The normalized write can store one value exactly, while key overlap gives an explicit interference law. Sparse activity alone guarantees neither unlimited capacity nor retention across changing worlds.

Self-modeling robots learn body models through action and sensation [11]; Polyworld evolves circuits with local Hebbian learning and metabolic costs [84]; PathNet evolves pathways trained by backpropagation [21]. Cadence’s organizing constraint is a decentralized model of computation: bounded observer-like patches retain state, exchange activations through declared ports and expose their readback. Useful behaviour must be built from these interfaces, with explicit teaching signals and memory writes.

Here *local repair* has an operational meaning. During inference, neuron i updates its state from its equation residual, $r_i = \sum_j W_{ij} \text{act}(v_j) + c_i - v_i$. Agreement means satisfying coupled constraints; different patches can publish different values. Synaptic learning changes these constraints using endpoint contrasts between perturbed equilibria. The earlier record mechanism instead corrects a witnessed prediction error at sparse addresses. Settling at fixed parameters does not itself learn. Local relaxation, equilibrium contrast and delta-rule memory are established ingredients [32, 70, 86]; each implemented combination needs its own convergence, credit and retention evidence.

1.3 The hypothesis

The hypothesis is that a reusable developmental procedure can produce networks that acquire, retain and transfer skills through experience across unfamiliar bodies and tasks. Fast activity carries the current interpretation; persistent learned relationships change what future equilibria can express. Perturbations from observations, teaching and consequences induce local parameter changes. The same bounded patch primitive should support perception, recall, action and the acquisition of useful distinctions, with specialized roles developing through experience and inherited rules rather than being assigned by the experimenter.

On this account the individual is the continuing organization of activity, parameters, interfaces and body coupling. A scalar energy or a currently stable output does not describe that organization completely. Contradiction can require revising an interpretation, changing a relationship, or taking an action that obtains discriminating evidence. A global workspace is one possible organization of these interactions, not an additional required controller or a conclusion about consciousness.

For composition, the actor is this whole continuing network. An intended phrase, recent sound, proposed motor command and predicted consequence must influence the same decision through declared ports. A current-frame predictor cannot establish that capability. Self-readback means exposing selected internal predictions or proposed actions to other patches so their consequences can be checked; simply stacking such interfaces proves no cognitive property. Neural and developmental organization supplies inspiration for simplified mechanisms, while causal intervention and task performance determine whether a particular abstraction is useful. A supplied brief cue can test retention of an intention without demonstrating autonomous intention formation.

A successful test must hold task-specific wiring out of the design process and count sensing, memory, communication, search and every learning phase, including any pretrained components. Sample efficiency, per-decision latency and total energy are distinct measures. The simulated-body experiments establish parts of the experience protocol; physical transfer, durable retention and lower total cost at matched competence determine whether the architectural thesis succeeds.

1.4 Technical contributions

Section 2 specifies the revised library interface and its interpretation boundaries. Section 3 evaluates the memory mechanisms that motivate its design. The earlier implementation and application evidence supply four further contributions, each retained with its original model assumptions.

- **A certified recurrent core.** For the implemented neuron, an explicit incoming-weight bound gives a unique equilibrium, a residual error certificate, and bounds on the repair required after changed input or

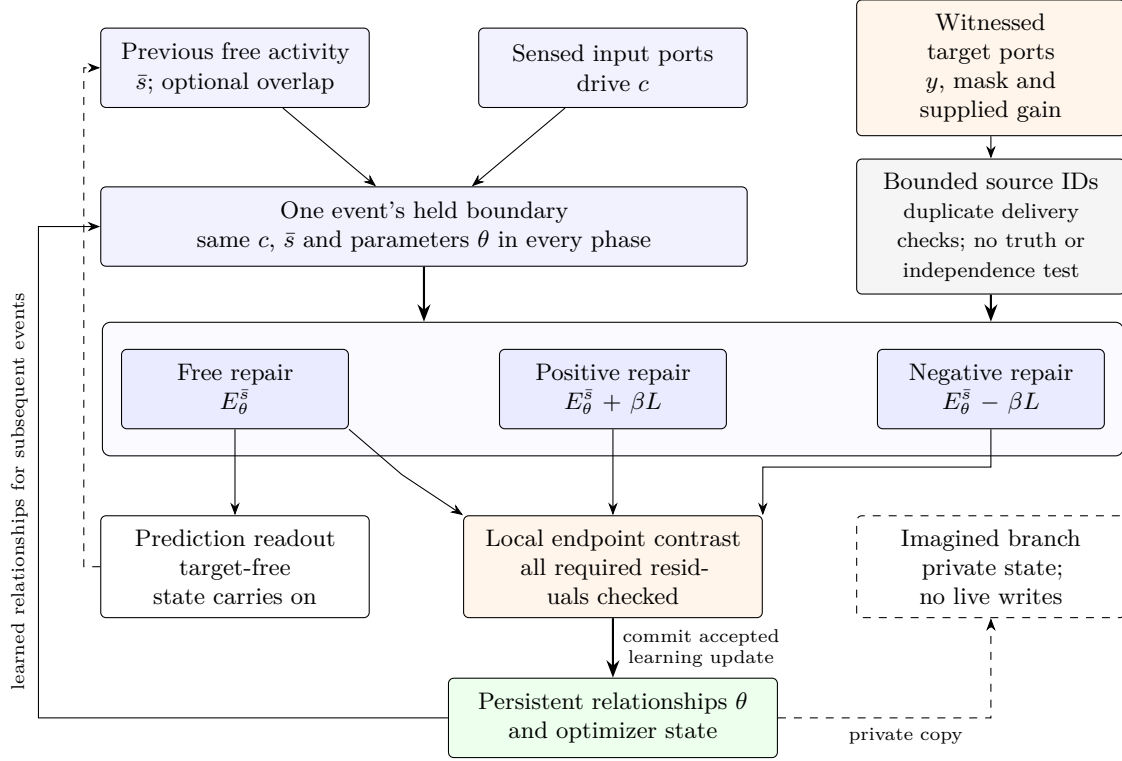


Figure 1: The revised core’s computational and evidence boundaries. Bounded neural patches repair local equations under common held inputs; only witnessed targets induce the default centred free/positive/negative learning contrast. The optional temporal overlap holds previous free activity fixed across those phases. Continuous learned relationships and activity persist, with different reset semantics. Source identifiers control duplicate delivery within a bounded window. Imagined branches evolve private copies without changing live parameters, activity or evidence bookkeeping. There is no separate record-memory cortex in this core.

lesions (Theorems 4.6, 4.3, 4.4 and 4.5). These connect local parameters to checkable network behaviour.

- **Writable memory with quantified interference.** The sparse record interface supports exact correction at a normalized key with unit learning rate and an exact law for how that write changes other reads. Separation and capacity bounds state when experience can be stored without overwriting another association (Theorems 4.9–4.12).
- **Structural change with a retained certificate.** Pruning, masking and gain reduction preserve contraction; added projections preserve it within a declared incoming-weight budget (Theorem 4.14). This supplies a mathematically admissible search space for recurrent wiring. Empirical genome searches have their own conditions and do not establish useful behaviour for every admitted graph.
- **An executable experience protocol.** Action, consequence readback, record writes and local learning form one continuing process. Candidate drives query records without writing imagined outcomes as witnessed facts. A neuron-by-neuron reference engine, accelerated implementations and source-bound receipts connect the computational rules to the reported applications (§4.10, §7).

Decentralization here describes state ownership and update dependencies. The software schedules phases, reduces residuals and coordinates record reads and action selection centrally; sparse-key selection also reads a group of cells. The experiments do not demonstrate an asynchronous deployment across machines. The separately evaluated combined learner includes a conventional gradient component. Distributing the complete controller requires an implementation and its own communication and latency measurements.

1.5 What the evidence establishes

The evidence concerns distinct systems. The revised core specifies a continuing interaction interface around the existing nonlinear recurrence and local learner. The small memory diagnostics test particular candidate mechanisms. Earlier application results concern closed-loop learning in the arm and artist (§7), useful record-bank allocation (§8.2), and conditionally cheaper numerical repair after localized changes (§7.4). The mathematical results in Section 4 state the guarantees of the certified subclass and record mechanism. Their declarations and receipts are mapped in the appendices.

The larger application brains are wired by hand. Allocation and evolution search within supplied families of mechanisms. The principal record-memory experiments supply addresses or clocks; conventional lookup and sparse solvers share several advantages. The arm’s records learner is slower per decision than its small multilayer control, and conventional sequence learners lead on text and music. A revised interface cannot inherit these applications’ scores before they have been run through it. The embodied-AGI argument is therefore an architectural hypothesis with multiple independent obligations, including physical-robot evaluation.

2 The revised core: one continuing patch network

2.1 Local state, learned relationships and ports

The library’s **PatchNet** interface wraps the existing nonlinear **Brain** and reciprocal **Learner** [53]. It owns a continuing free neural state, learned synaptic efficacies and biases, and the learner’s optimizer state. Input and output ports name the coordinates through which observations and predictions cross its boundary. A caller supplies their meaning, the initial wiring and the teaching objective. Internal functional specialization is a learning question; declaring a region to be a workspace does not establish that it performs a useful workspace function.

For the fast state, the concrete local constraint is

$$v_i = \sum_j W_{ij} \text{act}(v_j) + c_i,$$

with the nonlinear activation and repair step specified in Section 4. A patch reads its neighbours’ published values and its own boundary drive, then updates its discrepancy. Effective weights, rather than neuron labels or equal neighbouring values, define the relationship being repaired. The software schedules these local dependencies centrally. An energy interpretation requires the effective reciprocal interactions and the activation assumptions of the learning derivation; arbitrary directed connectomes do not acquire that interpretation by sharing the interface.

The retained computational state has the form

$$S_t = (v_t, \theta_t, o_t, P_t),$$

where v_t is free activity, θ_t the learned relationships, o_t the optimizer state and P_t a bounded optional record of delivered source identifiers. Fixed interface and solver configuration accompany this state. A complete reproducible system also accounts for any external random generator, body state and scheduler that affect subsequent observations. The optional identifier record provides duplicate-delivery bookkeeping; it is not an episodic content store.

The core does not attach the sparse records cortex used by the earlier applications. It also does not binarize synapses or add a separate importance classifier. Continuous learned weights can express intermediate relationships that a fixed binary-contact prior would eliminate. This is a smaller integration of existing Cadence mechanisms, with a specific empirical question: which useful memories can its activity and parameters actually acquire, preserve and revise?

2.2 An explicit temporal overlap

Carrying an initial state cannot preserve its information when complete settlement has one solution independent of that state. The optional temporal boundary makes the previous free activity $\bar{s}$ an argument of the

Operation	State and evidence contract
create	Supplies a reciprocal initial graph and named input/output ports; these are inherited structure, not learned roles.
settle, read	Advances and reads target-free activity without changing learned parameters. Inputs are soft drives and receive recurrent feedback.
observe	Uses externally observed target coordinates and supplied gains; updates parameters only after all required phases converge. A rejected attempt can advance free activity.
imagine	Evolves a sequential private branch; neither live continuation state nor witnessed evidence is changed.
reset	Clears fast activity while retaining parameters, optimizer state and delivery identifiers.
save, load	Preserves continuation state; exact numerical continuation requires the same backend and precision.

Table 1: The revised core’s public interaction contract. These operations expose computation and evidence boundaries; they do not guarantee learned competence.

constraints themselves. With a supplied mask $a_i \in \{0, 1\}$ and strength $\kappa \geq 0$, its energy term is

$$E_{\bar{\theta}}^{\bar{s}}(s) = E_{\theta}(s) + \frac{\kappa}{2} \sum_i a_i (s_i - \bar{s}_i)^2.$$

The corresponding local recurrent equation adds $\kappa a_i (\bar{s}_i - s_i)$ to the drive. The same $\bar{s}$ is held throughout the free, positive and negative phases of one observed event. It is not advanced at each numerical iteration and is independent of the target’s perturbation strength and evidence gain. The next event uses the preceding free activity; a fresh or reset stream uses a zero boundary. An imagined branch advances only its private previous state.

The default strength is zero, and the default eligible coordinates are the inherited hidden ports. Positive strength is therefore a supplied persistence assumption. It introduces no additional fact store and reuses existing activity, but it changes the model’s equations and must be counted as a structural choice. The unmodified recurrence’s contraction certificate does not automatically certify this changed solver. In a single-minimum landscape, repeated temporal overlap can merely slow forgetting. It does not discover a protected subspace, choose important events or guarantee indefinite retention. The gradient estimate below is conditional on the held boundary; it does not differentiate through preceding events or solve long-delay temporal credit.

An exact scalar example distinguishes persistence from protection. For $a > 0$ and $\kappa \geq 0$, the energy $az^2/2 + \kappa(z - c)^2/2$ has unique minimizer qc , where $q = \kappa/(a + \kappa) \in [0, 1]$. Reusing that minimizer as the next boundary gives $q^n c$, tending to zero. These statements are machine-checked [Lean: `repair_unique`, `iterate_closed_form`, `iterate_tends_to_zero`]

. They describe a convex scalar system; they do not prove that every nonlinear neural basin washes out.

A readable trace is a different boundary. The library also provides **Trace**, used by earlier controllers but not automatically instantiated by **PatchNet**. With movement weighting disabled, it records selected free activity h_t as

$$z_{t+1} = \rho z_t + (1 - \rho) h_t, \quad 0 \leq \rho < 1,$$

and makes z_t readable at declared context ports during the next event. The trace is held fixed across all numerical iterations and nudged phases, then advanced once from the free state. This supplies an explicit fading observation history; carrying a numerical initial state or tethering the immediately preceding activity implements a different operation.

For fixed z_t, h_t , the displayed update is the unique minimizer in u of

$$\frac{\rho}{2}(u - z_t)^2 + \frac{1 - \rho}{2}(u - h_t)^2.$$

Six machine-checked lemmas establish this scalar repair, interval preservation, decay of an initial difference as ρ^n under the same source sequence, and decay to zero under zero source [Lean: `unique_minimizer`, `repair_mem_interval`, `same_source_history_difference`, `zero_source_tends_to_zero`]

. Here the timescale and source are supplied. If the recurrent source itself depends on the trace, the same-source premise need not hold. These lemmas certify neither a joint energy for the full recurrent system nor its stability, learned importance or permanent memory [R80].

2.3 Readback and detuning as the learning operation

A witnessed target y and an observed-coordinate mask m define a port loss, for example

$$L(s; y, m) = \frac{1}{2} \sum_k m_k (s_k - y_k)^2.$$

Unobserved coordinates do not supply targets. At fixed parameters and fixed temporal boundary, the default centred learner obtains a free state and states under positive and negative perturbations $E_\theta^s \pm \beta L$. It then estimates a parameter derivative from the difference of the corresponding local energy derivatives:

$$g_e(\beta) = \frac{\partial_{\theta_e} E_\theta(s^+) - \partial_{\theta_e} E_\theta(s^-)}{2\beta}.$$

For the implemented pair interaction, this is an endpoint activity contrast. The current optimizer and parameter constraints determine the ensuing update. The gradient interpretation is imported from equilibrium propagation under its smooth stable-branch, symmetry and convergence assumptions [42, 70]; endpoint locality alone proves none of those assumptions. Finite nudges are biased, branches can differ in a multibasin system, and free-state convergence does not imply convergence of both nudged phases.

The revised interface checks the equation residual of every required phase and refuses a parameter update when a phase reaches its work cap without meeting tolerance. It reports convergence and work rather than treating small successive movement as sufficient evidence of settlement. This is a numerical acceptance contract. Outside the contractive subclass it is neither a bound on distance to a unique equilibrium nor a proof of local stability or correct task predictions.

The distinction between activity repair and learning is essential. A free equilibrium answers the current constraints. A witnessed teaching perturbation changes persistent relationships only through the learning operation. An unobserved read does not supply a fabricated target. The caller’s loss, target and evidence selection therefore determine what counts as a useful correction; a harmful-outcome nudge is supplied valence, not autonomous discovery of importance.

2.4 Memory, imagination and evidence identity

Free activity carries forward to the next decision. Resetting fast activity leaves learned relationships intact, allowing a cold query to test parameter memory separately from a lingering response. Checkpoints preserve parameters, optimizer state, activity and bounded delivery bookkeeping so that restart is not confused with relearning. These contracts make retained state inspectable; task-specific retention must be measured.

Imagined alternatives settle from isolated copies of the free state. They cannot update the live state, learned parameters or witnessed-event bookkeeping. Their predictions can guide an external action, but agreement among simulated branches is not fresh evidence about the world. A branch that predicts a shock and an instrument that reads back a shock have different evidential roles. Learning from a generated target would require an explicitly different, disclosed training protocol.

This boundary is enforced by the branch operation’s lack of writes. The caller supplies observations to the learning interface; the library does not authenticate that a caller-provided target came from a physical measurement. Source identifiers authenticate neither that provenance nor the target’s truth.

An optional source identifier suppresses repeat delivery of the same observed payload while that identifier is retained, and rejects conflicting reuse of that identifier. Finite storage limits this guarantee. Distinct identifiers do not establish independent evidence; repeated reports from one sensor can share a cause. The core does not construct an authenticated causal graph, infer sensor reliability or decide whether a contradiction is noise or a changed world.

2.5 What an equilibrium can remember

Fast memory and parameter memory differ. At fixed drive and parameters, global contraction implies one equilibrium. Consequently complete settlement discards differences carried only by the initial activity. Warm starts save work but do not evade this implication. History can instead affect a persistent parameter, an explicit boundary value, or a basin choice outside global contraction. A multiple-basin system can preserve a cue under unchanged observations; learning changes its landscape and can destroy that cue while preserving the old cold input-response mapping.

Sufficiency concerns future behaviour. Let h be a history, T a declared deterministic update, and r its required readout. The full future signature is

$$F(h)(u) = r(T^*(h, u)), \quad U(F, i)(u) = F(i :: u), \quad r_F(F) = F(\square),$$

where u ranges over finite input continuations. Histories with identical signatures admit the same shift update and present readout. Every exact memory with state-only updates and exact readout must distinguish histories with different signatures. These statements are machine-checked [Lean: `future_signature_one_step`, `every_exact_dynamic_encoding_refines_future_signature`]

- . This is a behavioural quotient, not a finite learned compression algorithm: the signature can be infinite and a callable representation may retain the whole history. A family of N behaviourally distinguishable histories requires at least N distinguishable memory states [Lean: `distinguishable_family_capacity_bound`]

- . Finite-precision parameters therefore have a capacity limit; counting real coordinates without precision does not remove it. The required task and continuation family determine which distinctions matter. The construction does not reveal arbitrary future importance to a learner [54].

A causal poset supplies dependence, not missing content. Two records with different values can have exactly the same causal order and maximal node identifiers. The corresponding topology-only encoding fails any query that distinguishes those values [Lean: `topology_only_is_insufficient`, `maximal_ids_only_are_insufficient`]

- . A sufficient frontier must retain the required payloads or boundary summaries as well as causal context. Disjoint replacement writes commute under their update law; conflicting writes on one key do not [Lean: `distinct_key_updates_commute`, `same_key_updates_do_not_commute`]

- . Provenance can therefore support safe delivery and revision, but a DAG alone provides neither storage semantics, learned importance nor confluence.

Normal form is not objective truth. A terminating, confluent repair relation gives a unique normal form from a starting state under its hypotheses. A boundary-determined answer additionally requires all relevant history to be represented in that boundary. Neither condition implies that noisy observations were interpreted correctly. In a two-world construction, the same available reports require latent answers zero and one. Any single estimate z then satisfies

$$z^2 + (z - 1)^2 \geq \frac{1}{2},$$

so at least one squared error is at least $1/4$ [Lean: `two_target_squared_errors`, `indistinguishable_noise_or_change_forces_error`]

- . This is an information bound, not a stochastic noise model. Immediate perfect revision and perfect immunity to an indistinguishable first outlier cannot both be required. Acquiring another informative observation is one possible response to that uncertainty.

Conditional basin retention. Let B be a supplied basin, let an allowed exit from B raise a reference energy by at least $m > 0$, and suppose each time-varying energy E_t differs from that same reference by at most ϵ at every state under consideration. If $2\epsilon < m$, an allowed step that does not increase E_t cannot leave B . Every finite accepted trace therefore stays in B , preserving any readout constant there [Lean: `finite_trace_preserves_basin`, `finite_trace_preserves_readout`]

- . A per-step bound relative only to the preceding energy is insufficient for this conclusion. The theorem assumes the exit margin and allowed moves; it does not certify the numerical neural implementation's basins, find their margins, or prove retention under arbitrary future learning.

Specialization needs asymmetry and useful consequences. A deterministic permutation-equivariant update preserves an invariant initial state under invariant inputs. Exchangeable owners with identical complete states and exposures cannot differentiate by that update alone [Lean: `equivariant_rollout_stays_fixed`, `homogeneous_owner_rollout`]

. Random initial conditions, unequal experience or inherited asymmetry can remove this obstruction. They do not themselves establish useful division of labour. A symmetric task can have asymmetric optima. For a supplied mismatch r and connection price $c > 0$, minimizing $w(r^2 + c)$ over a freely adjustable weight $w \geq 0$ selects $w = 0$ [Lean: `positive_cost_selects_zero`]

. Retaining useful communication therefore needs a task constraint or consequence beyond cheaper agreement.

3 Memory diagnostics and the architectural hypothesis

The experiments distinguish direct library tests from separately specified mechanism candidates. Their common purpose is to test which distinctions a local repair system preserves, which observations change them, and whether the changed state supports useful subsequent behaviour [54].

A direct library test. A separate test of `PatchNet` without an explicit temporal boundary trains four supplied indicator cues with continuous targets, presenting two old associations and then two new ones. Across three seeds, all end-of-stage acquisition errors lie below the declared threshold of 0.01, but only 800 of 960 attempted updates are committed. One seed rejects every update in the second stage because its nudged phases do not converge; its new-association error of 0.00893 was present before that stage. Threshold satisfaction therefore does not demonstrate new learning in that seed. Two seeds finish below the old-association error threshold; the third finishes at 0.02924. Resetting fast state after initial acquisition preserves the learned response. However, after a cue followed by one blank settlement, none of the trained graphs preserves the declared 0.1 output distinction: the largest cue-dependent difference is below 3×10^{-6} . A separately supplied two-neuron reciprocal latch does retain activity history; its wiring is inherited, not learned. Checkpoint continuation, imagined-branch isolation and bounded duplicate-delivery checks pass. These are developmental measurements on a small supplied representation, not evidence of learned general memory [R75].

Temporal overlap changes the tradeoff. A fixed $\kappa = 1$ comparison uses the same three seeds and commits all 320 attempted learning updates per seed after converged phases. Cue-dependent output differences after the first blank settlement are 0.16577, 0.12326 and 0.16886, all above the supplied distinction threshold. By the sixteenth blank settlement all are below 5.4×10^{-5} . The change provides finite context in these graphs, not permanent activity memory. Cold acquisition after the first stage has mean-squared errors 0.01163, 0.02149 and 0.01643; after the second stage, new-association errors range from 0.02067 to 0.02860. Every one misses the 0.01 acquisition criterion. A later low error on an old association cannot establish successful retention when its initial cold acquisition failed. This comparison isolates a useful temporal boundary while exposing its interaction with cold retrieval and learning; neither tested setting meets the joint memory requirements [R76].

A post hoc retrieval diagnostic starts from the same final checkpoints and repeats the real cue for sixteen events on private branches, with no target or learning update. New-association errors then fall below 6×10^{-8} in all three seeds; one seed’s old-association error is 0.02384. Thus some information inaccessible to immediate cold readout can be recovered by continued cue presentation, while interference persists. This selected diagnostic does not replace the original cold-query criterion or establish retention through blank input [R76].

The library receipts include complete saved endpoints, checkpoints and immutable source copies. Verification independently recomputes the saved equation residuals and observed errors; full training and branch isolation are reproduced with the archived implementation. This distinction matters: replay supports reproducibility, while the verifier does not independently derive every training update. The mathematical theorems do not replace these implementation checks [R75,R76].

Selective plasticity can preserve learned facts. A controlled extension separates retrieval, input feedback and shared-parameter interference in a thirteen-neuron network with four indicator input ports. Teaching

first supplies two old targets, then two new targets, then changes one old target. A cold probe resets fast state before each cue. The per-cue squared-error criterion is 0.01; successful retention requires prior acquisition. Omitting a temporal anchor when no preceding state exists restores cold acquisition in five fresh initializations, but none preserves every untouched fact after revision. Fixing the observed input coordinates, including inactive coordinates at zero, eliminates changes to inactive input contacts but also gives zero complete passes. Shared hidden and output relationships can change the responses to other cues even when their input contacts do not change [R78].

A calibrated comparison gives both arms the same adequate readout capacity, fixes input coordinates, and uses the existing equilibrium-detuning update. One arm learns all relationships; the other permits changes only on input-incident contacts, with shared hidden/output relationships and biases held fixed. At 160 updates per teaching stage, the localized arm passes every acquisition, retention, revision and convergence condition in two of five initializations. Its other failures are insufficient acquisition within the exposure budget. At 640 updates per stage on five fresh initializations, it passes all five; the matched all-plastic arm passes none. Every one of the 19,200 scheduled updates commits after converged phases. In the localized arm, earlier cold responses are exactly unchanged by unrelated teaching and the three untouched responses are exactly unchanged by revision [R78].

The noninterference is conditional and transparent. Every changed input contact has a presynaptic publication of zero on an unrelated cue. Since $\text{act}(0) = 0$, those changes contribute zero to every free coordinate’s transport. Reciprocal changes ending at fixed input coordinates also leave the free equations unchanged. With unchanged shared parameters, the same boundary and the same numerical initial state therefore produce the same constrained repair. This argument does not cover overlapping addresses or unequal temporal anchors. The indicator addresses, calibrated readout and plasticity mask are supplied; no useful specialization or importance is discovered.

Both calibrated arms use the same graph and teaching exposure, but the localized confirmation costs 1,102,496 integration row-steps versus 170,496, a factor of 6.47. Each numeric checkpoint state occupies 5,309 bytes; including metadata and the bounded delivery ledger raises the serialized array payload to about 185 kB. An independent verifier checks saved equation residuals, fixed coordinates, exact protected parameter values, cold-readout invariance and replayed blank phases. It does not independently replay every optimization step. Five initializations of one supplied four-fact world do not establish indefinite retention, noisy addressing or compositional generalization [R78].

A real-audio feature test. A fixed-seed Amen diagnostic applies the same core to 64 continuous descriptors from recorded audio, using inherited ports and separate held-out excerpts. Both $\kappa = 0$ and $\kappa = 1$ models commit 190 learning updates. One-step old/new mean-squared errors are 0.09490/0.08187 without temporal overlap and 0.09495/0.08210 with it; a fourth-order ridge autoregression scores 0.05950/0.04693. For sixteen-step continuation, the overlap model scores 0.09603/0.07947, versus 0.08400/0.07342 for a first-order autoregression. Both neural configurations reach an exactly unchanging output tail in the numerical rollout despite reported convergence. Independent metric arithmetic and source checks support these comparisons; the audio verifier does not replay all neural endpoints. This is a feature-prediction test, with no motor controller, rendered composition or claim about musical quality. The additional uploader-linked mix is treated as a distinct recording, without assuming alignment to another video’s timestamps [R77].

A scalar periodic diagnostic separates missing observations from failed acquisition. Across three initializations, current-only, temporal-overlap and explicit preceding-observation arms all lose autonomous tail variation. The explicit-history arm also has teacher-driven mean-squared error 0.02570, while a ridge model with the same two observations has autonomous error 1.14×10^{-8} . A supplied-clock neural control retains variation but misses the joint accuracy and amplitude criterion. Thus these failures cannot be attributed solely to an absent previous frame; the tested learner has not acquired a sufficiently accurate transition [R79].

Retained intention must influence action. A separate whole-network diagnostic connects eleven input ports, eight hidden neurons and two output ports. One brief cue asks for ascending or descending four-note motifs after a shared four-event neutral prefix. The net receives no continuing goal label, phase or sequence index. Its two outputs propose a motor command and predict the next heard value. The supplied instrument returns the clipped motor as sound, making the two teaching targets identical; this is not an independent test of a learned body model. Matched arms read either no trace or the existing eight-coordinate **Trace**, with

decay 0.9. Imagined trajectories use private copies and never supply teaching targets [R81].

All six models, three initializations per arm, converge throughout 2,048 training events, but neither arm acquires the required autonomous motifs at either measured budget. Mean primary error is 0.01933 with the trace and 0.018754 without it, against 0.01875 for constant output. Under identical heard inputs through the first intended action difference, the trace retains between-intention distances of approximately 0.003–0.012. Swapping only the trace changes that motor output by less than 0.0005, against a required contrast above 0.2; the no-trace control is unchanged. Information is retained and weakly causal, but not used for the required control [R81].

Training deterministically alternates intentions and carries state continuously, whereas primary evaluation starts cold. Those confounds prevent attributing the failure to one memory or credit mechanism. A post-run diagnostic replays demonstrations as heard input without learning. Starting from the exact saved state gives error 0.017662, versus 0.017647 for constant output on that training cycle; cold replay gives 0.017848. Removing or reversing subsequent cues changes little. The failure is therefore not explained entirely by cold initialization. Post-run linear probes can separate intentions within the continuous training history but fail to transfer their decision thresholds reliably to cold traces; they do not establish an acquired neural readout. The intent experiment’s independent verifier checks source custody, reported metrics, trace updates, instrument feedback and behavioral criteria, without replaying every neural solve or auditing the exploratory linear decoder [R81].

A fresh episodic comparison removes those two confounds: intentions are sampled independently and activity and trace reset at every seventeen-event episode. With the same graph and 2,048 learning events, both ordinary arms fail in all three fresh initializations. Mean errors are 0.019065 with trace and 0.019074 without it. An acquisition control instead supplies at every event an indicator address for each intention–motif–position combination and signed intention. This engineered representation lets the same neural graph and local update acquire the targets in all three initializations, with mean error 0.00004461. All 18,432 training events commit and all checked endpoints converge. The control imports the temporal organization that the actor must learn; its eight test rows repeat four distinct intention/primer conditions because cue strength is unused. It establishes conditional acquisition capacity, not learned timing, intention or a composer [R82].

Candidate and measurement	Positive observation	Decisive boundary
Nonlinear basin memory; five trained lineages, two cues each [R71]	All ten carried cues survive blank or irrelevant cell observations.	Every isolated opposing cue replaces the old cue at every tested gain. During compatible new-task learning, four old carried cues change despite preserved cold-query skills.
Nine learned scalar patches; five lineages [R72]	Held-out masked completion is 99.46%, and all 120 initial memories hold.	Only 13 of 120 episodes both retain the initial memory and reach the correct revised state. All 60 necessary conjunction revisions leave that field stale.
Nonlinear contact stabilization; supplied input geometry [R73]	A strong isolated association persists through 64 idle repairs while a weak isolated trace decays.	A representable shared-connection task needs an intermediate weight; the binary-contact prior prevents learning both associations. Strong false observations are stabilized too.

Table 2: Separate mechanism diagnostics, including their failed conditions. Completion coordinates and memory episodes share trained lineages and are not independent trained models. Numerical convergence, learning quality and retention are distinct measurements.

Persistence without reliable revision. The basin-memory stress test retains all scheduled trials and reports endpoints at a 64-event horizon. The persistent deep regime holds ten of ten cues through blank and irrelevant observations, whereas reset and shallow controls score five of ten. Under compatible subsequent learning, six carried cues are correct and four change. Fresh queries to both old and new learned cue-response mappings are ten of ten correct. Thus the four failures concern information held by activity, rather than demonstrated erasure of the old learned skill. All four erroneous final states meet the numerical residual

tolerance; solver completion does not fix their content [R71].

Useful learned relationships without general specialization. The nine-patch diagnostic trains on static masked worlds with redundant views of two bits and their conjunction, with no group label supplied to the learner. All world values occur in training; held-out masks test completion. Learned edges score 99.46% across pooled eligible coordinates, compared with 75.95% for tied edges and 75.59% after shuffling learned edges. The primary scores include capped numerical endpoints. The warm revision test instead starts from an established memory. Its final endpoints converge and have positive local Hessian curvature, yet the learned model leaves the conjunction stale whenever it must change. This establishes a training-to-execution mismatch in the tested system, not an impossibility for its energy family. Lesion patterns suggest localized dependence, but do not establish general developmental specialization [R72].

A structural-memory rule and its cost. A separate contact experiment applies

$$R(p) = 3p^2 - 2p^3, \quad 0 \leq p \leq 1,$$

after a local witnessed update. This is descent for the supplied double-well potential $V(p) = p^2(1 - p)^2$: $R(p) = p - \frac{1}{2}V'(p)$. For each $0 \leq c < 1/2$, define $q = c(3 - 2c) < 1$. Lean proves

$$0 \leq p \leq c \implies 0 \leq R^n(p) \leq q^n p,$$

and the reflected bound $1 - R^n(p) \leq q^n(1 - p)$ for $1 - c \leq p \leq 1$, together with energy descent on the unit interval [Lean: `lower_basin_geometric`, `upper_basin_geometric`, `repair_decreases_potential`]

. These bounds apply during unforced repair. They do not select salience or protect against later conflicting writes.

The prediction-port experiment supplies stronger teaching gain to designated events. Its strong one-event retention therefore does not demonstrate learned importance. The more decisive failure uses $x_A = (1, 0)$ and $x_B = (1/2, \sqrt{3}/2)$ with both targets equal to one. The allowed weights can represent both exactly at $p = (1, 1/\sqrt{3})$. After learning A and then repeatedly observing B , the structural rule stabilizes $p = (1, 0)$, preserving A but predicting only $1/2$ for B . A no-leak continuous-weight control learns B while retaining an approximate A . Individual stable contacts do not necessarily preserve a useful distributed relationship. The revised core consequently keeps continuous relationships without installing this contact prior [R73].

Exploration must obtain useful evidence. Passive settlement cannot resolve distinctions absent from its inputs. An exact finite reference task provides three noisy cues, a costly diagnostic action, and an equally novel but irrelevant coin observation. Selective inspection has expected utility $297/320 = 0.928125$, compared with $9/10$ for always inspecting and $27/32$ for passive majority. These values follow exhaustive enumeration under the supplied observation model. They define an evaluation instrument; they are not a trained Cadence policy or a discovery of curiosity [R74]. A neural test must learn both when inspection is worthwhile and when to stop, using the same repair and detuning mechanism without receiving an oracle uncertainty label.

These findings motivate continuous learned relationships, explicit sensory evidence and tests that combine acquisition, hold, revision and further learning. They do not motivate adding every tested control as a new subsystem. Preserving a rare useful fact, discarding an irrelevant one and correcting a strong false belief jointly require information about future consequences that the primitive alone does not supply.

4 The certified recurrent subclass and sparse records

This section retains the mathematical account of the earlier recurrent and record implementations. The revised core uses the same neural recurrence, but its contractive theorems apply only when their hypotheses are checked. The sparse-record results concern an optional distinct memory mechanism, absent from the minimal revised interface. The later application sections measure the specific combinations documented by their receipts.

Computational choice	Mechanism evaluated here	Theorem	Measured in
Fixed work per input	work follows the change; an unchanged input needs no repair step	4.4	§7.4
Context and cached activations	potentials, traces and records persist across decisions	4.9	§7.3
Attention to a growing history	a record read and write whose cost is independent of how much has happened	4.13, 4.15	§7.3, §6
Append-only context tokens	one-shot write, exact revision, no interference between separated keys	4.9, 4.12	§7.3
Weights freeze before use	one mode; the same synapses change during use	4.7	§7
Backpropagation through stored iterations	each synapse reads its own two endpoints and one broadcast error	4.7	§7.5
Interference on tested streams	a sparse code confines each update to the records the reading touched	4.12, 4.10	§7
Recomputed hypothetical inputs	alternatives are settled from the current state and scored by records	4.4	§4.6
Damage changes the model	displacement after a lesion is bounded; the same local rule repairs it	4.5	§7.6
The architecture is chosen by hand	the wiring is a genome; budgeted mutations preserve convergence	4.14	§8

Table 3: Properties of the earlier contractive and sparse-record implementations. These results do not certify the revised interface’s task competence or multibasin retention. The theorems are machine-checked in Lean (Appendix A); measurements have hashed receipts (Appendix B).

4.1 Neurons, synapses, settling

A net has n neurons. Neuron i holds a potential $v_i \in \mathbb{R}$ and publishes an activation $s_i = \text{act}(v_i)$. Synapses carry effective weights W_{ij} from neuron j to neuron i ; in the library the effective weight is the product of a gain, the number of contacts, a learned efficacy and a presynaptic factor, and the paper writes the product as W_{ij} . A stimulus c_i is an additive drive, with the bias absorbed into it. One settling step is

$$v_i \leftarrow (1 - dt) v_i + dt \left(\sum_j W_{ij} \text{act}(v_j) + c_i \right), \quad 0 < dt \leq 1. \quad (1)$$

The activation is a rectified, re-based logistic. With slope λ , threshold θ , leak $\ell \in [0, 1]$ and $\sigma(x) = 1/(1 + e^{-x})$,

$$\rho = \sigma(-\lambda\theta), \quad r(v) = \sigma(\lambda(v - \theta)) - \rho, \quad \text{act}(v) = \frac{\max(r(v), 0)}{1 - \rho} + \ell \frac{\min(r(v), 0)}{\rho}. \quad (2)$$

Rest publishes exactly zero, the activation rises to one, and a neuron below rest publishes a small negative value so that the learning rule can move it. The learning model uses $\lambda = 1$, $\theta = 0$, $\ell = 1/10$. A settling run stops when the largest activation movement in a step falls below a tolerance or when a step cap is reached, and the library reports the residual of equation (1) at the final state. Adaptation, a slow variable subtracted from the drive, produces rhythmic dynamics; the theorems below are stated without it.

The whole brain is one such net. Regions are named ranges of neurons with declared projections between them: senses, association, a workspace that carries context, motor slots, a critic. Settling is joint over all of them, which is what lets a partial reading be completed from the rest of the state.

Intuition: each neuron checks its own account. Imagine an arm whose visual input has moved while its motor and body predictions describe the preceding position. Each neuron compares its held potential with the drive its neighbours and senses supply. Equation (1) moves the potential partway toward that drive; the published activation changes what neighbouring neurons receive, and they make the same comparison. The

resulting computation is a feedback process with a measurable discrepancy at every step. In the observer-patch description, each bounded patch owns state, communicates through declared ports, reads back its consequences and retains records. A settling certificate and the public evidence bundle make that self-reading process checkable. The equilibrium reconciles the equations the model implements; whether those equations describe the arm well is a separate empirical question.

4.2 From observable events to a global normal form

OPH distinguishes internal descriptions from shared readings [57–59]. A visual patch may locate an object in camera coordinates and a motor patch in body coordinates. Agreement concerns the position after translation to a shared interface. A change of internal coordinates and corresponding interface map is a gauge change when it preserves every declared public reading. Gauge-invariant repair corrects disagreement in those readings.

For local states x_i and readout maps $\pi_{i,e}$ into a common interface on an overlap $e = \{i, j\}$, agreement means

$$\pi_{i,e}(x_i) = \pi_{j,e}(x_j).$$

A repair preserves the observations held fixed. OPH’s local-repair criterion requires every allowed fixed exterior to admit a consistent interior. A *normal form* has no repair left to make. For a terminating discrete relation on the observable quotient, local confluence gives one normal form from each starting state. Completeness, meaning that normal states coincide with consistent states, makes it a global solution. Local compatibility alone need not imply global consistency. For different interiors with the same protected observations to yield the same answer, the compatible consistent states must also form one equivalence class [58, 59].

What counts as an observable event? For a finite classical record, events are sets of possible readings, closed under union, intersection and complement. OPH’s verified projection-event calculus supplies conditional expectations that discard internal distinctions while preserving the probabilities of the declared partition events [55]. Cadence uses classical activation ports and writable records; the quantum projection and conditioning machinery is mathematical background, without a claimed implementation in this neural model.

The neural specialization. Hold the weights fixed, protect $B(c, v) = c$, and define consistency by $v = W \text{act}(v) + c$. Under Cadence’s contraction condition, every initial state at a fixed input approaches the same equilibrium. The residual and stimulus bounds quantify accuracy and sensitivity. This realizes an observation-determined equilibrium: local equations determine one global answer with a checkable error bound. Numerical convergence replaces finite exact termination; the synchronous proof does not establish arbitrary asynchronous or transactional repair guarantees. Learning determines whether the answer is useful.

4.3 The certificate: how much work a decision needs

Write $F_c(v)$ for the right-hand side of (1) at stimulus c , $\|\cdot\|_\infty$ for the sup norm on $\mathbb{R}^n$, $\rho_W = \max_i \sum_j |W_{ij}|$ for the largest absolute incoming weight sum (the row mass), and L for a Lipschitz constant of act .

Theorem 4.1 (settling certificate). *If $0 < dt \leq 1$ and $L\rho_W < 1$, then F_c is a contraction of $(\mathbb{R}^n, \|\cdot\|_\infty)$ with constant $q = 1 - dt(1 - L\rho_W) < 1$.* [Lean: `settle_contracting`]

Corollary 4.2 (unique equilibrium). *Under the hypotheses of Theorem 4.1 the net has exactly one equilibrium v_c^* , and every fixed point of F_c equals it; with zero stimulus and $\text{act}(0) = 0$ that equilibrium is rest.* [Lean: `equilibrium_unique, silence`]

Theorem 4.3 (a priori and a posteriori certificates). *Under the hypotheses of Theorem 4.1, for every start v and every k ,*

$$\|F_c^k(v) - v_c^*\|_\infty \leq \frac{q^k}{1 - q} \|v - F_c(v)\|_\infty, \quad \|F_c^k(v) - v_c^*\|_\infty \leq \frac{\|F_c^k(v) - F_c^{k+1}(v)\|_\infty}{1 - q}.$$

[Lean: `apriori_bound, aposteriori_bound`]

A residual below ϵ at the returned state certifies a potential error of at most $\epsilon/(1-q)$, and an activation error of at most L times that. This is an error bound relative to the model's equilibrium, not a certificate of prediction accuracy. Banach's theorem is the whole proof [6]; what the paper adds is the exact condition on this neuron (Theorem 4.6) and the receipt that the bound never failed in 57,600 checks (§7.4).

Theorem 4.4 (the equilibrium is Lipschitz in the stimulus, and warm starts pay). *Let F_c and $F_{c'}$ satisfy Theorem 4.1 with the same constant q and let $\delta = \|c - c'\|_\infty$. Then*

$$\|v_c^* - v_{c'}^*\|_\infty \leq \frac{dt \delta}{1-q}, \quad \|F_{c'}^k(v_c^*) - v_{c'}^*\|_\infty \leq \frac{dt \delta q^k}{1-q} \quad \text{for every } k.$$

[Lean: equilibrium_lipschitz_stimulus, warm_start_bound]

This is the theorem behind adaptive work. Starting from the old equilibrium, the number of steps that reaches accuracy ϵ after an input change δ is at most the nonnegative ceiling of $\log(dt \delta / ((1-q)\epsilon)) / \log(1/q)$: logarithmic in the size of the change, zero when the change is zero. The bound is sufficient rather than a prediction of semantic difficulty, and checking a returned state has its own cost.

Intuition: reuse the answer, then bound the repair. Contraction says that two possible internal states move closer under the same input: after one step their largest difference is at most q times its previous value. With $q = 1/2$, every step halves that uncertainty. If the residual at a returned state is 10^{-3} , the remaining potential error is at most $2 \cdot 10^{-3}$; the factor $1/(1-q)$ sums all the smaller corrections that can follow. When a sensor changes slightly, the preceding equilibrium is a useful starting point because Theorem 4.4 bounds how far the required answer can move. This can save repeated work on a slowly changing stream. It gives little advantage after a large change, and convergence can be expensive when q is close to one. A unique equilibrium also erases differences in starting state: lasting history must enter through records, traces or a changed drive, rather than through the warm start alone.

Theorem 4.5 (bounded damage). *Let $|\text{act}| \leq A$, let S be a set of neurons, let W^S be W with the columns in S set to zero, and let $\mu_S = \max_i \sum_{j \in S} |W_{ij}|$. If F_c satisfies Theorem 4.1, then the lesioned map satisfies it with the same constant, and $\|v^* - v^{*,S}\|_\infty \leq dt \mu_S A / (1-q)$. [Lean: lesion_bound, lesion_contracting]*

Theorem 4.6 (the library's neuron). *For $\lambda \geq 0$ and $\ell \geq 0$, the activation (2) is Lipschitz with constant $\frac{\lambda}{4} \max(\frac{1}{1-\rho}, \frac{\ell}{\rho})$, satisfies $|\text{act}| \leq \max(1, \ell)$, and publishes exactly zero at rest. Under the learning model the constant is $1/2$ and the bound is 1, so every net whose row mass is below 2 settles by a certified contraction for every $0 < dt \leq 1$. [Lean: activation_lipschitz, activation_abs_le, activation_zero, learningActivation_lipschitz, learning_model_certified]*

For the hypothesis, the certificate makes the numerical state of a changing brain checkable: when its conditions hold, a returned decision carries a bound on its distance from equilibrium. This bound certifies the solve. It does not certify the accuracy of the learned body model or the safety of an action.

4.4 Learning by two equilibria

The free phase settles under the stimulus alone to s^0 . The positive nudged phase settles from s^0 with an extra drive of strength β on the output neurons toward a target: for a quadratic nudge the drive is $\beta(\text{target}_i - s_i)$, for a cross-entropy nudge $\beta(\text{target}_i - p_i)$ with p the softmax of the outputs at temperature T , one softmax per declared slot. The centred estimator settles a negative nudged phase at $-\beta$ from the same free state. With s^+ and s^- the two nudged equilibria, the contrast at synapse (i, j) and at neuron i is

$$\Delta_{ij} = \frac{1}{B} \sum_b \frac{s_{b,i}^+ s_{b,j}^+ - s_{b,i}^- s_{b,j}^-}{2\beta}, \quad \Delta_i = \frac{1}{B} \sum_b \frac{s_{b,i}^+ - s_{b,i}^-}{2\beta}. \quad (3)$$

Reciprocal synapses share one parameter and receive the mean of their two contrasts; a cap bounds every efficacy.

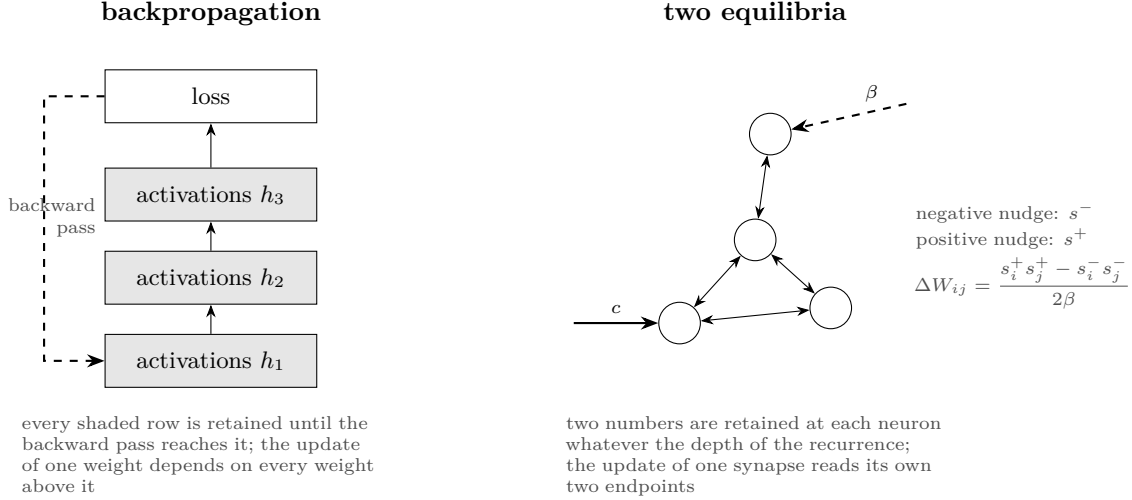


Figure 2: What an update must know. Backpropagation retains the forward tape and propagates a global signal backward through it. The contrast of two equilibria retains two numbers per neuron, and each synapse computes its own change from its two endpoints (Lemma 4.7). The two agree to four digits on the same mechanism (§7.5).

Lemma 4.7 (locality). *There is a fixed function g of four real numbers such that the contrast of synapse (i, j) equals $g(s_i^+, s_j^+, s_i^-, s_j^-)$; the contrast is symmetric in i and j ; and if the two compared equilibria agree, every contrast is zero. [Lean: `contrast_local`, `contrast_symm`, `contrast_eq_zero_of_agree`]*

The lemma describes the information used by the update (Figure 2). Reverse-mode differentiation retains or recomputes intermediate activations; the equilibrium contrast instead reads each synapse’s endpoints in the perturbed states. This can reduce storage when differentiating through many settling iterations. Both perturbed equilibria must be computed, and implicit differentiation offers another way to avoid retaining the iteration history [5].

Fact 4.8 (the contrast is the gradient). *On a smooth stable equilibrium branch with symmetric effective recurrent weights, converged phases and no adaptation, the limit $\beta \rightarrow 0$ of the contrast is minus the gradient of the nudge’s loss with respect to the weight of a reciprocal pair [70]; the centred estimator cancels the first-order bias in β [42]. The library’s tests check the identity numerically: at $\beta = 10^{-4}$ and phase residuals below 10^{-11} the contrast equals the finite-difference gradient to a relative tolerance of 10^{-5} .*

Free and nudged phases are the deterministic form of the two phases of the Boltzmann machine [1, 52], equivalent to backpropagation in a layered net [83], and the local-activation-difference rule of generalized recirculation [61]. Reward uses the three-factor form: an eligibility trace $e \leftarrow \gamma \lambda e + \Delta$ of the same contrast, a linear critic with its own trace, and one broadcast error $\delta = r + \gamma V(s') - V(s)$, with the update $\eta \delta e$ [22, 73, 76].

Intuition: let a small correction travel through the circuit. The target gently pushes the outputs, and recurrence transmits that change to the rest of the net. A synapse compares how strongly its two endpoints coactivate under the positive and negative pushes. Their difference tells it which direction the connection should move under the assumptions of Fact 4.8. Credit reaches the synapse through the circuit’s changed state, so the update itself needs only its endpoints. There is a cost: the learner must solve the perturbed equilibria, and a locally computed update can be slower than a conventional backward pass. An eligibility trace carries that local comparison forward when reward arrives later; it does not guarantee that one delayed scalar identifies the right cause. The advantage is the form and storage of the update, with learning quality established by the controlled comparisons.

4.5 Records: what followed each reading

A record cortex keeps, for each of its cells, what followed the readings that cell took part in. A reading $x \in \mathbb{R}^d$ of declared source neurons is made mean-free by a running mean $\bar{x}$, projected by a fixed random matrix R onto $D \gg d$ cells, rectified, and reduced to the a strongest entries at unit length,

$$k(x) = \text{kwta}_a(\max(R(x - \bar{x}), 0)). \quad (4)$$

Each predicted field holds one table $M \in \mathbb{R}^{D \times m}$. The prediction is the read $k(x)M$, and the witnessed outcome y is written by the delta rule [38, 81] at rate η ,

$$M \leftarrow M + \eta k(x)^\top (y - k(x)M). \quad (5)$$

Consequence fields use a slow rate so that a regularity is averaged over many outcomes; valued fields such as reward use rate one, so that one exposure writes. Figure 3 shows the parts.

Theorem 4.9 (one-shot memory and its interference law). *For any record M , keys k, q , value y and rate η over a commutative ring: (i) the read at q after the write changes by $\eta(q \cdot k)$ times the correction $y - kM$; (ii) if $k \cdot k = 1$, the read at k after a write at rate one is exactly y ; (iii) if $q \cdot k = 0$, the read at q is unchanged; (iv) if $k \cdot k = 1$, writing the same correct observation twice at rate one changes nothing.* [Lean: `read_write`, `read_write_self`, `read_write_orth`, `repeat_write_stable`]

Theorem 4.10 (how far one write reaches). *Over the reals, the change a write at k makes to the read at q has sup norm exactly $|\eta| |q \cdot k|$ times the sup norm of the correction, and an overlap of at most ϵ bounds it by $|\eta| \epsilon$ times that norm.* [Lean: `read_write_dist`, `read_write_dist_le`]

Theorem 4.11 (exact storage and capacity). *After writing N pairwise orthogonal unit keys in sequence at rate one, every one of them reads its own value, whatever the initial record was; and over a field at most d keys can be pairwise orthogonal unit vectors in a d -dimensional key space.* [Lean: `read_writeUpTo`, `capacity_le`]

Theorem 4.12 (pattern separation). *Two keys whose supports are disjoint are orthogonal, so a write at one leaves the read at the other exactly unchanged; a family of unit keys with pairwise disjoint supports, written in sequence at rate one, is stored exactly from any initial record; and a code of width D with a active cells admits at most $\lfloor D/a \rfloor$ such keys.* [Lean: `dotProduct_eq_zero_of_disjoint_support`, `read_write_of_disjoint`, `read_writeUpTo_of_disjoint`, `disjoint_family_card_le`]

Intuition: a correction follows the overlap of two addresses. Suppose a reading activates two cells equally, giving a unit key $k = (1, 1, 0, 0)/\sqrt{2}$. A write at rate one corrects that reading in a single step. A second reading with key $q = (0, 0, 1, 1)/\sqrt{2}$ shares no cells and keeps its prediction exactly. A third, $q = (0, 1, 1, 0)/\sqrt{2}$, shares one cell and receives half the first reading's correction, since $q \cdot k = 1/2$. This is the interference law in a small example: the same overlap that permits generalisation can overwrite another fact.

The four theorems explain why a useful code can learn quickly from a stream: each observation directly corrects the relevant prediction while disturbing other predictions in proportion to their key overlap. Sparsity alone guarantees neither distinct addresses nor retention; two sparse keys can be identical, and finite capacity limits exact separation. With an appropriate code, direct writes can need fewer observations than fitting the same associations in distributed weights. The record theorem does not by itself prove a replay requirement for another architecture. Section 7 measures that comparison on the declared tasks.

Theorem 4.13 (attention is one read of a record). *Let $M = \sum_t k_t^\top y_t$ be the additive record of N key/value pairs. The read at query q is $qM = \sum_t (q \cdot k_t) y_t$, the stored values weighted by the query/key dot products. If the keys are one-hot at distinct addresses, the read at a one-hot query is exactly the stored value, for every N up to the size of the address space.* [Lean: `read_hebb`, `one_hot_recall`]

The first sentence is unnormalised linear attention [3, 36, 71]; the softmax read of a transformer differs by an exponential kernel and a normaliser, and modern Hopfield networks show that it is one step of an energy network [39, 40, 66]. The residual write of equation (5) is the delta rule over a sequence [86]. The identity covers additive dot-product attention, not the general softmax operation. Unlike an additive write, the residual write corrects its current prediction.

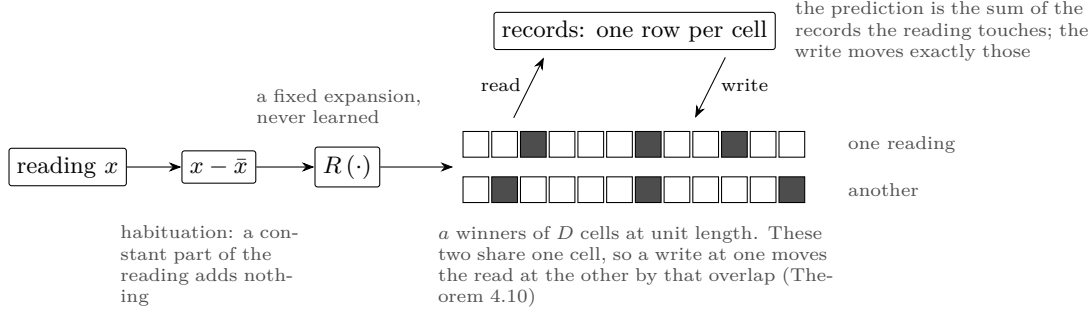


Figure 3: The records cortex. The nonlinearity lives in the code and the learned map is linear and local, the arrangement of the cerebellar granule layer [2, 47] and of the dentate gyrus [48, 62]. Overlap of codes decides interference, and the winner-take-all keeps the overlap small.

4.6 Symmetry breaking: thinking before acting

A settled state is a fixed point for a particular drive. To consider an alternative, the brain can add a small random drive or encode a candidate action in the input, hold that changed condition during the imagined solve, and settle from the preceding equilibrium. Theorem 4.4 bounds the work after a change in additive drive when the contraction conditions hold. A transient perturbation of state alone returns to the same equilibrium under an unchanged contractive map; it cannot select a second basin. An imagined action produces an imagined reading, whose consequences are read from the records, and a search over such readings chooses what to do. Reading a record changes nothing, so imagination leaves no trace in what the brain has learned; and the imagined state is composed by the same machinery that composes the witnessed one, so a brain can imagine only what its own records support. The arm imagines 364,284 futures per life and the artist about a million, none of which is written [R51,R54]; the Connect Four brain imagines boards through its own learned drop records and searches them with alpha-beta pruning [R53].

4.7 A genome for the wiring

The parts above are fixed within a life. What is open is how they are wired: which fields a record cortex reads, at what gain and offset, how many cells and winners it has, how many pathways each cell samples, which regions project to which, and how many cortices there are. In the library that description is a genome. A genome develops into a brain at a seed, a brain lives one life, and a mutation of the genome changes region sizes, projection densities, signs and gains, adds or removes a cortex, flips a bit of a reading mask, or splits one cortex into two of half the size. Evolution is a loop over genomes with a fitness read from short lives; nothing acquired in a life passes to the next. The rule that the code is fixed within a life is deliberate: the records are addressed by the code, and a code that changed under a life would misaddress everything written before.

Theorem 4.14 (the certificate under evolution). *Let W satisfy Theorem 4.1 with row mass ρ . (i) Every W' with $|W'_{ij}| \leq |W_{ij}|$ for all i, j satisfies it with the same constant: pruning a synapse, masking a projection, lowering a gain and scaling by any $|g| \leq 1$ are instances. (ii) If V has row mass μ and $L(\rho + \mu) < 1$, then $W + V$ settles by a certified contraction with constant $1 - dt(1 - L(\rho + \mu))$: growth by new projections stays certified inside the row-mass budget. (iii) Every member of a population whose genomes share one row-mass bound settles by one certificate, whatever its individual wiring. [Lean: rowMassLE_of_abs_le, rowMassLE_smul, rowMassLE_add, mutation_contracting, growth_contracting, population_contracting]*

The theorem permits structural search within a checked row-mass budget. It certifies convergence for admitted recurrent weights, not useful behaviour or every empirical genome. Section 8 reports structural searches with their own measurement conditions.

Intuition: search for a useful view of the experience. A record head that reads pitch, gate and unrelated sensory fields together can give similar codes to actions whose consequences differ. Splitting that reading into appropriate heads can make the consequence predictable and protect it from unrelated writes.

The genome searches over such choices; experience fills the resulting records. The row-mass budget limits how strongly recurrent connections can amplify one another, so a mutation admitted within that budget preserves convergence. It supplies no test of whether the chosen sensors or divisions are useful. That test comes from held-out predictions or survival under the world’s resource costs. The wiring experiments therefore distinguish finding a useful layout within a supplied family from inventing an unrestricted architecture.

4.8 Cost

Fix the declared cost model. A patch net with E synapses that settles in k steps costs kE multiply-adds per decision, plus Dm for a record read and the same for a write, and holds n potentials and Dm numbers whatever has happened before. A transformer with ℓ layers of width d over a window of T tokens costs about $\ell(12d^2 + 2Td)$ multiply-adds per token and holds a key/value cache of $2\ell Td$ numbers.

Theorem 4.15 (what grows and what does not). *Under that model, the arithmetic a transformer spends on a stream of T tokens is $\ell(12d^2T + dT(T - 1))$, which grows quadratically in T , and its cache grows without bound. The arithmetic a patch net spends is $T(kE + 2Dm)$ and its state is constant in T . For every choice of positive ℓ, d, k, E, D, m there is a horizon beyond which the patch net’s total is the smaller of the two. [Lean: `attention_total`, `attention_total_ge_quadratic`, `record_total`, `record_state_const`, `exists_crossover`]*

The theorem is arithmetic about the declared model, not a claim that the two objects compute the same function. It states where the shapes differ: one of them re-reads its past, the other has written its past into a fixed store. Section 6 puts measured rows beside it.

Intuition: compression trades history for a bounded state. For a repeated body-control task, a learner may need the present position and a learned map from actions to consequences, rather than every observation that established the map. Records and traces carry that summary between decisions. Their fixed size explains the linear stream cost in the theorem, and also imposes a limit: they cannot preserve arbitrarily many independent facts exactly. The quadratic comparison assumes attention to an expanding history; a transformer with a fixed window also has linear total cost in stream length. Whether retaining the summary is better depends on what the task requires, the cost of finding the representation, the achieved error and the measured latency.

4.9 What the theorems tell a designer

Require $L\rho_W < 1$ and $0 < dt \leq 1$ when using the contraction certificate; the learning model has row-mass limit two (Theorem 4.6). Check the potential residual against $\epsilon(1 - q)$ and report both the error bound and the work spent (Theorem 4.3). Warm-start every stream and account for the change in drive (Theorem 4.4). Separate keys before writing, and measure the overlap after separation: disjoint supports guarantee noninterference, sparse expansion alone reduces it (Theorems 4.10 and 4.12). Give the consequences a slow rate and the reward a fast one. Let a search prune, mask and rescale freely and charge growth against one row-mass budget (Theorem 4.14). Distinguish the lesion bound on potentials from measured recovery under further learning (Theorem 4.5).

4.10 Receipts

Every experiment writes a receipt: canonical JSON whose digest covers the kind, the body and a manifest of source hashes, so that editing any source file invalidates it. A reference engine settles neuron by neuron with one message per declared synapse and a ledger of transmissions, and the accelerated engines are checked against it to $2 \cdot 10^{-16}$ on the CPU. The applications additionally ship a verifier that recomputes every published number from the event logs, and a browser port of the same brain that is checked decision by decision against the Python one. The receipts this paper cites are copied into its evidence package with their hashes (Appendix B).

In the animal	Described by	The part here	Measured in
Granule layer: a few inputs per cell, Golgi inhibition, a sparse expansion read by one Purkinje cell	Marr [47]; Albus [2]; Litwin-Kumar et al. [45]; Cayco-Gajic and Silver [12]	the fixed random projection with winner-take-all of equation (4)	§7.3, §7
Climbing-fibre error at the cell that made the prediction	Ito [33]; Raymond and Medina [68]	the delta write of equation (5), one error per field per cell	§7.3
Dentate gyrus and CA3: conjunctive codes whose overlap bounds interference	Marr [48]; Treves and Rolls [77]; O'Reilly and McClelland [62]; Leutgeb et al. [44]	pattern separation before the write	Thm 4.12, §7.3
Fast hippocampal store beside slow cortical structure	McClelland et al. [50]; Kumaran et al. [41]	two rates: slow for consequences, one exposure for reward	§7
Persistent activity as working memory	Goldman-Rakic [28]; Wang [79]	the carried equilibrium, the afterglow and the traces of §5	§7.3, Appendix D
Adaptation to a constant input	Barak et al. [7] on coding level	the running mean subtracted before the expansion	§7
Recurrent settling that completes a partial input	Hopfield [31, 32]; Rao and Ballard [67]; Friston [24]; Bastos et al. [8]	the joint equilibrium of the settled regions	§7.4
Error-driven learning from local activation differences	O'Reilly [61]; Ackley et al. [1]; Scellier and Bengio [70]; Laborieux et al. [42]	the free and nudged contrast of equation (3)	§7.5
Dopamine reports a reward prediction error on a trace left by recent coactivity	Schultz et al. [73]; Yagishita et al. [85]; Frémaux and Gerstner [22]; Gerstner et al. [26]	the three-factor rule and its eligibility trace	§7
Place cells and a map that survives a change of goal	O'Keefe and Dostrovsky [60]; Hafting et al. [29]; Stachenfeld et al. [75]; Whittington et al. [80]	records keyed by place; a goal changes the reward field alone	§7
Place sequences that depict paths before the animal moves	Pfeiffer and Foster [63]; Matar and Daw [49]	imagined futures from a broken symmetry, searched and discarded	§4.6, §7
A global workspace that broadcasts to every specialist	Baars [4]; Dehaene et al. [17]; Dehaene and Naccache [16]	the jointly settled workspace in the world and composer applications	Appendix D, §11
Synapses that hold state over many timescales	Fusi et al. [25]; Benna and Fusi [10]; Clopath et al. [13]; Frey and Morris [23]	fast and slow strengths with consolidation	§10
Plasticity that does not switch off	Dohare et al. [18]; Zenke and Laborieux [87]	one mode of operation; no frozen checkpoint	§7
Wiring inherited across generations, knowledge acquired within a life	the order of every nervous system	the genome of §4.7; records reset at birth	§8
Measured wiring diagrams	Cook et al. [14]; Dorkenwald et al. [19]; Knox et al. [37]	a connectome loads as the synapse table of the same net	§9

Table 4: Each part of the design, the biological or algorithmic motivation, and the section that measures it.

5 Biological motivations and their implemented counterparts

Biological theories motivate several parts of Cadence, without making it a biophysical model. This section describes the mechanisms of the earlier applications, including optional traces and records absent from the revised core. Table 4 gives the limited correspondences, and Figure 4 shows two application networks while they work. A biological analogy neither identifies a mathematical energy with a brain's objective nor demonstrates that the same functions emerge from an unstructured population.

One stream, and what carries across a moment. The applications learn during an ongoing interaction. Their retained state has three forms: settled potentials, traces and records. Warm starts can save computation when the drive changes little (Theorem 4.4), but convergence to a unique equilibrium erases the initial state

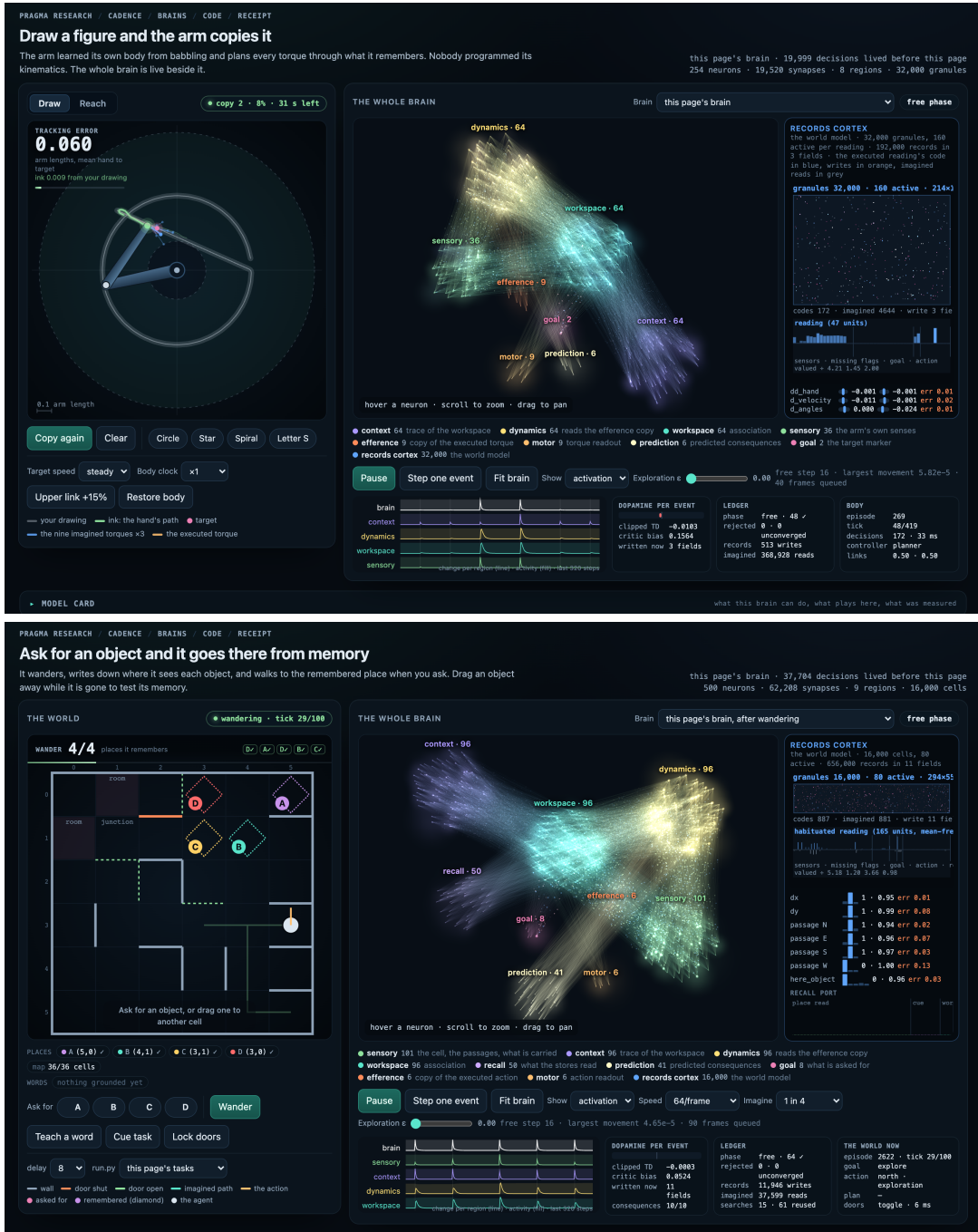


Figure 4: Two of the four lives, captured from the archived interactive demonstrations while they run; every panel displays the network’s own state. Top: the arm copies a circle a visitor drew, at a tracking error of 0.060 arm lengths, after settling nine imagined torques per decision. The connectome shows eight regions in their colours, brightness for activation, a glow where neurons change, and lit synapses carrying messages; the free phase settled in 16 steps. The right panel is the records cortex, 32,000 granules with 160 active for the executed reading, and the predicted consequences beside their witnessed values, after 368,928 imagined reads and 513 writes. The change per region over the last 320 steps, below the brain, is a display of where activation changed; these traces alone do not show which arithmetic was skipped. Bottom: the world brain wanders a map it has never seen, writes down where it sees each object, and walks there from memory when asked; the ledger shows 11,946 writes and 37,599 imagined reads with no rejected update, and the dopamine panel the clipped reward error of the last event. Both brains keep learning while the visitor plays with them.

at a fixed drive. Unlike persistent-activity accounts of working memory [28, 79], this contraction mechanism therefore needs traces, records or another change to the effective input to retain history.

Short-term memory: the afterglow and the traces. The second kind of carried state is written by no synapse. An afterglow is a trace of a sensory region that decays geometrically after each event, so a moving object leaves a fading streak in the retina’s copy and a motion is readable from one frame; the player of Appendix D has an afterimage of 16,992 neurons beside a retina of the same size. A working-memory trace is the same operation on a cortex rather than a sense: the composer’s prefrontal region is a trace of its phrase cortex with decay 0.85, and its piece region a slower trace with decay 0.98, both supplying retained stimulus to the next equilibrium and read by the phrase cortex through learned synapses [R68]. Fast synapses are the third form, a delta-rule record written every event and decaying, which is what holds a digit span of 12 from position tags [R5]. These explicit stores need no gradient through time. Learning long dependencies through a recurrence can face vanishing or exploding gradients [9, 30]; that is a credit problem, not an absence of recurrent memory. Transformers also retain information in their context and cache. Table 7 tests the particular alternatives on delayed cues.

Long-term memory. Complementary-learning-systems theories distinguish rapid episodic acquisition from slower accumulation of shared structure [41, 50]. In the earlier applications, the fast store is the records cortex: a reading lights a sparse code, and what followed is written into exactly those cells by the delta rule of equation (5), in one shot at rate one for a valued field and at a slow rate for a regularity. Sparse conjunctive coding in the dentate gyrus and CA3 keeps the overlap between two episodes small so that one can be stored without overwriting another [44, 62, 77]; Theorem 4.10 is that argument in numbers and Theorem 4.12 its limit. The slow store is every plastic synapse of the settled cortices, changed by the two-equilibria contrast of equation (3) and by the three-factor rule when a reward error arrives. Both stores are written during use and neither is frozen. Section 10 records that the slow store is weaker on the tested task families.

Attention goes to the repair. A local change can leave much of a recurrent equilibrium nearly unchanged. Theorem 4.4 bounds displacement; it does not supply a scheduler or guarantee sparse arithmetic. Figure 4 displays activation changes. A separate event-driven numerical prototype schedules the affected work and checks the full residual: on the fly graph it answers a local drive change in 5.43 ms against 34.49 ms for full iteration (§7.4). This saving is conditional on locality and scheduling overhead.

Completion, imagination, one scalar for reward, habituation. Recurrent networks that settle into a state which completes a partial input are Hopfield’s construction [31, 32], and predictive coding gives feedback connections the role of carrying a prediction and feedforward connections the role of carrying what it missed [8, 24, 67]; here the joint equilibrium of the settled regions completes a reading with a missing field. Before a rat runs to a remembered goal, place-cell sequences depict the path, including paths never taken [49, 63]; the counterpart is the symmetry-broken settling of Section 4.6. Dopamine neurons report a reward prediction error [73] and convert a recent coactivity flag into a lasting change only within a window of about one second [85], the eligibility trace of three-factor theory [22, 26]; the library broadcasts one such scalar into a local trace at every plastic synapse. Adaptation to a constant input [7] is the running mean subtracted before the expansion, so a constant part of a reading adds nothing to the code.

The expansion and the local readout. A granule cell receives about four mossy-fibre inputs, five orders of magnitude fewer than a Purkinje cell receives parallel fibres, and Golgi feedback keeps the code sparse [45, 47]; Albus [2] cast the circuit as a perceptron whose expansion recoding raises discrimination and learning speed, and the same motif appears in the dentate gyrus [12] and in the fly’s mushroom body [15]. With a fixed expansion and squared-error linear readout the fitting problem is convex, and each update touches only active cells. Stability and retention depend on the step size, key overlap and changing targets. The lives of Section 7 measure whether this memory is useful without replay; they do not prove that another learner must replay.

The global workspace. Global workspace theory holds that a set of specialised processors compete for access to a shared workspace whose contents are broadcast back to all of them, and that this broadcast is what makes a content conscious [4]; Dehaene et al. [17] gave it a neuronal form as a network of long-range neurons that ignites when a content wins the competition, and Dehaene and Naccache [16] set it out as a framework for the neuroscience of conscious access. The hand-wired world and composer brains include a workspace region, a cortex that receives projections from the specialist cortices and projects back, settled jointly with all of them, whose trace is the brain’s context; in the world’s brain of Figure 4 it is the 96-neuron region at the centre. The paper makes no claim about consciousness. It does hold that the workspace is where the hypothesis needs the decision to be made, and Appendix D reports that in the composer the planner reads the records around it.

What the design does not have. Dendritic subunits that compute conjunctions before learning [27, 43, 65] have no counterpart; the neuron is single-compartment. Offline prioritised replay [49, 82] exists in the library as an optional ring and is switched off in every application of Section 7. Metaplastic chains that trade capacity for retention [10, 25] are present only as two timescales.

6 When retained state saves work in the measured systems

A transformer cache avoids recomputing earlier keys and values, although ordinary attention reads a history that grows with context. A fixed-size recurrent state and record bank instead summarize that history at bounded storage cost, trading explicit access to old tokens for finite memory capacity. Theorem 4.15 gives a crossover in its declared arithmetic model, not a universal wall-clock advantage over attention or other bounded-state learners.

Warm starts provide a second possible saving: the required repair shrinks with the input change. A sparse scheduler can exploit localized changes, as the separate solver experiment in Section 7.4 demonstrates. Full residual audits, queue management, initialization and dense changes remain costs. The application brains do not all implement that scheduler.

The learning comparison is likewise conditional. Local contrast can save the tape used to differentiate through settling, but pays for free and perturbed phases. A record can absorb an association in one update, whereas a gradient learner may need repeated samples. Table 7 measures this on particular streams, counting the controls’ replay; the table also reports the resulting latency disadvantage.

Table 5 collects every row of the paper in which a patch net and a transformer, or the best standard sequence learner, are measured on shared datasets or matched interaction protocols. Arithmetic is not wall-clock, and the same table says where the implementation loses: a patch-net decision on a CPU costs 12.9 ms in the arm’s life against 1.0 ms for a multilayer network in the same life, because settling is sequential and the implementation is a small dense kernel, and on text the conventional sequence learner wins outright.

The receipts report CPU costs of 2,849 seconds for an arm life, 1,933 for the world and about 7,375 for the artist suite, with peak memory of 105 to 115 megabytes; the Connect Four life with its deeper search costs more, at a decision latency of 201 ms at the 95th percentile [R51,R52,R53,R54]. The arm holds 254 neurons with 19,520 directed synapses and 192,000 numbers of records; the world 500 neurons, 62,208 synapses and 656,000 record numbers; Connect Four 187 neurons, 3,960 synapses, 14,000 drop records and 6,000 line records; the artist 511 neurons, 52,160 synapses and 2,240,000 record numbers. No accelerator is used, no corpus is read, and the whole training set is the stream the brain lived through.

7 Earlier applications: one stream, one life

7.1 Four lives

Four applications test learning during interaction. Each primary life begins with random synapses and empty records, with no pretraining or replay ring. Frozen evaluations and born-frozen controls separately measure generalization and the contribution of learning. The brain receives moments and returns decisions; the world, its rules and its rewards sit outside it. Receipts bind source hashes, event logs and checkpoints, and verifiers recompute the results. Acceptance uses five fresh seeds at full budget.

task	patch net	transformer or best sequence learner
recall over 128 pairs	1.00, 0 trained parameters	0.01, 108,224 parameters, 20,000 steps
revision stream, 128 writes	1.000, 576 bytes per stream	0.145, 17,960 parameters, window of 17,408 bytes
arm, held-out targets, one stream	0.994, no stored transitions	0.402 with 102,790 parameters; 1.000 with a ring of 4,096
world, cue across eight steps	1.000	0.516 with a transformer, 0.588 with a multilayer network
four task families, 5,000 decisions	0.946 with in-life allocation	0.490 online transformer; 0.663 with a ring
four task families, 320,000 decisions	0.981	0.822 online transformer; 0.836 with a ring; 0.866 gradient net with a ring
digit span	12 from 5,696 parameters	12 from 53,770; 5 from 80,138 (LSTM)
three-back, four-back	0.964, 0.954 from 2,744	1.000 from 51,458 with the answer in the window; 0.910, 0.791 from 20,098
cart-pole, steps to threshold	40k from 1,716 parameters, 10.3k multiply-adds per decision	137k from 17,443, 67.8k multiply-adds
decision after an unchanged input	0 repair steps plus a checking step	the full stack
permuted MNIST, mean of thirty	0.913 from 80,339	0.840 from 10,986 (patch transformer)
button calibration	0.455 bits per press	0.806 (GRU), 1.500 (convolutional net)
music, held-out surprise, 65,188 pieces	1.023 from 52.8 million parameters	0.708 from 7.6 million (GRU)
character text, bits per character	3.29 at width 128	3.03 at width 128, 2.67 on validation at epoch 5

Table 5: Every matched row in the paper, wins and losses. Text and music favour the conventional sequence learner. Parameter savings in the memory rows coexist with supplied addresses and with exact dictionary ties (§10).

7.2 Learning during the life, from one stream

The comparison runners use the same world protocol, sensor encoding, planner, decision budget and five seeds, and replace the consequence learner: the records cortex, an online multilayer network, or an online transformer, each with and without a replay ring of 4,096 stored transitions that supplies one replayed update per real one [R55]. These are separate closed-loop lives: different actions produce different later observations. Table 7 reports adapting held-out evaluations, not frozen generalization on an identical trajectory.

The replay ring substantially improves both conventional models on reaching. The records learner achieves comparable performance without storing transitions, but uses 192,000 record values against the MLP’s 2,182 parameters, and costs 12.9 times its decision latency. The delayed-cue gap also depends on the supplied representation and memory access. The interference identity describes record updates; it is not a theorem about the controls’ replay requirements or their achievable capacity.

The held-out target score permits learning during evaluation and therefore measures adaptation to fresh targets. Freezing the learned arm lowers success from 0.994 to 0.953; the artist’s corresponding foreground score falls from 0.932 to 0.885 [R51,R54]. Both remain competent when frozen, while the adapting scores measure the full experience loop. These comparisons support useful online memory under a shared planner, without a general claim against gradient learning.

7.3 Owned state: one-shot memory and its limits

Memory inside a life. In the world, a request to fetch an object seen earlier from a cell half the map away succeeds 1.00 of the time; with the place records erased right before the request the same histories succeed 0.18 [R52]. When an object is moved in view, the memory is corrected 1.00 of the time. When the doors stop

Life	What it learns	What it reaches	Controls that fail
Arm [R51]	the consequences of its own torques, from motor babbling	0.994 of held-out targets reached while learning; 0.953 with learning frozen; copies a visitor's drawing at tracking error 0.052 against a bar of 0.113; a longer link is absorbed at 0.992 and the original body returns at 0.991	born frozen 0.000, random torques 0.002, shuffled pairing loses 0.224, corrupted dynamics loses 0.404
World [R52]	the consequences of moving, and where it saw what	consequence accuracy 0.979 on held-out episodes; requests from a distant cell fulfilled 1.00; the cue held across eight steps 1.00; a moved object corrected 1.00; doors that stop opening recovered 1.00; words grounded 1.00	memory erased 0.18, words erased 0.10, shuffled pairing gap 0.968, reward-shifted cue 0.514
Connect Four [R53]	what a dropped stone does, which windows are lines, and what its own searches proved, from its own games	exact next boards 1.00 on 10,000 held-out moves, terminal prediction 1.00, tactical suite 1.00, 0.9994 against random and 0.9946 against a one-ply opponent, twelve gates of twelve	corrupted dynamics loses 0.807
Artist [R54]	what its strokes leave on a canvas	held-out foreground 0.932 while learning, 0.885 frozen; a longer link mid-life 0.926 and the original body 0.927	born frozen 0.00, random scribbling 0.08, the supplied path oracle 0.64, an online multilayer network with a replay ring behind the same search 0.80, shifted canvas: 0.708 with readback, 0.436 without

Table 6: The four lives, five acceptance seeds each, every gate passed. None of them writes a replay ring: the receipts record zero replay writes.

	records	MLP	MLP, ring	transformer	transformer, ring
arm, held-out targets reached	0.994	0.018	0.980	0.402	1.000
arm, copying a drawing (error)	0.052	0.702	0.102	0.596	0.058
world, consequence accuracy	0.979	0.949	0.952	0.932	0.948
world, cue across eight steps	1.000	0.588	0.492	0.516	0.484
world, cue across 32 steps	1.000	0.576	0.486	0.494	0.466
parameters or records (arm)	192,000	2,182	2,182	102,790	102,790
decision latency, arm (ms)	12.9	1.0	1.0	5.6	8.5
stored transitions	0	0	4,096	0	4,096

Table 7: Matched life protocols with the world model replaced, five seeds; trajectories depend on each learner's actions. Held-out evaluations permit continued learning. Without a replay ring the multilayer network reaches 2 percent of the arm's held-out targets and the transformer 40 percent; with a ring they reach 98 and 100. The records brain reaches 99.4 percent from its own single stream with no ring. On the cue that has to be carried across a delay, every network arm sits near chance whatever its ring. Latency is the price: a decision costs 13 times the multilayer network's.

key cosine	keys	rounds	code cosine	plain record	separated	separated, uncentred	dictionary
0.0	64	4	0.014	0.512	1.000	1.000	1.000
0.5	64	4	0.014	0.325	1.000	0.725	1.000
0.9	16	4	0.014	0.200	1.000	0.500	1.000
0.9	64	4	0.014	0.138	1.000	0.350	1.000

Table 8: Correlated keys under revision, all-key accuracy, five seeds. Sixty-four keys at cosine 0.9 in a 32-dimensional key space, revised four times, are held exactly after separation and at 0.138 without it. The mean absolute cosine between codes is 0.014 whatever the cosine between keys, which is Theorem 4.10 in numbers. Without the running mean the shared component chooses the winners and the record fails. An exact dictionary is perfect here and is the control this row does not beat.

opening, the life recovers 1.00. A reward that follows a cue seen eight steps earlier is chosen 1.00, against 0.514 when the reward is shifted onto the other cue, which is the control that fails when a brain has memorised the room rather than the cue.

Associative recall and revision. A vocabulary of 128 symbols, a context of N key/value pairs, then a query. The net writes each pair as one outer product and answers by settling with the key as stimulus. No parameter is trained. Accuracy is 1.00 at every length from 4 to 128 pairs. A two-layer transformer of 108,224 parameters trained for 20,000 steps on contexts up to 32 pairs reaches 0.26 at four pairs and 0.01 at 128 [R1]; its training loss ended at 3.7 to 4.1 against 4.85 for a uniform prediction, so the row states what that budget buys rather than what attention can do. Under a stream that revises eight keys, the residual record holds 128 writes at 1.000 where a two-layer transformer trained on lengths 8 to 32 falls to 0.145 outside its training lengths and an additive record to 0.260 [R2]. A later token can correct an earlier statement in a context; a residual record instead overwrites its prediction directly (Theorem 4.9(ii)).

Correlated keys. The case where a plain record fails is the one pattern separation is for. Keys of dimension 32 with pairwise cosine up to 0.9, 16 or 64 of them, each revised over four rounds [R3]. Table 8 gives the receipt.

Span and clocked memory. Fast synapses from position tags reach a digit span of 12 with 5,696 parameters where an LSTM trained through time with Adam reaches 5 with 80,138, and a one-layer transformer over the trial reaches 12 with 53,770 [R5]. On n -back, records written from a period- n clock reach 0.943, 0.982, 0.964 and 0.954 at one to four back with 2,744 parameters, against an LSTM’s 1.000, 0.989, 0.910 and 0.791 with 20,098, and a windowed transformer whose window contains the answer at 1.000 with 51,458 [R6]. The position tags and the clock are supplied by the environment, which Section 10 counts as a debt.

7.4 Work follows the change

Measured. A layered net under the learning model, efficacies scaled to row masses of 0.5, 1.0 and 1.5, five seeds, 64 stimuli. Each stimulus is settled cold to a tolerance of 10^{-6} ; the input is then changed by a sup-norm amount δ and the settling restarts from the old equilibrium [R8].

Sparse repair on a connectome-sized graph. A separate numerical prototype uses the supplied adult fly graph, 138,639 neurons and 2,700,513 directed synapses, under a known contractive rule [19, 72]. The solver receives the graph and changed source identities; it does not learn them or predict the living animal. Each query starts from a common settled baseline. Across 135 conditions, every timed answer includes a full residual audit [R10]. A single-neuron drive change at tolerance 10^{-7} takes 5.43 ms against 34.49 ms for full iteration, a mean per-seed speedup of 6.30 with 10.4 times fewer synapse visits; 32 changed neurons give 5.84 times and 32 ablations 2.98 times. A change to every neuron makes the queue 2.6 times slower. On digits the settling count is 20 on clean inputs, 20 on noisy ones and 23 on occluded ones [R12]: displacement bounds do not measure semantic difficulty.

change δ	0	10^{-3}	10^{-2}	10^{-1}	$3 \cdot 10^{-1}$	1
warm steps to the new equilibrium	0.0	10.0	14.0	17.3	19.0	20.3
warm steps, the library’s stopping rule	1.0	9.0	13.0	17.0	18.0	20.0
the certificate’s budget	0	27	35	43	47	51
cold steps	21.0	21.0	21.0	21.0	21.0	21.0

Table 9: Compute follows change, row mass 1.0, five seeds. An unchanged input needs zero repair steps and one checking step; a thousandth of a change costs half a cold settling; the count grows with the logarithm of the change. Across the three row masses the a posteriori bound of Theorem 4.3 held in 57,600 of 57,600 checks and the displacement stayed inside the bound of Theorem 4.4 in every cell. A stack of K layers spends K layers in every one of these columns.

rule	new drives	stronger drives	eight lesions	retained state	fit seconds
centred contrast (two equilibria)	4.56865×10^{-8}	1.42443×10^{-7}	1.49987×10^{-7}	110,592 B	0.852
backpropagation through the solver	4.56913×10^{-8}	1.42457×10^{-7}	1.50013×10^{-7}	825,744 B	0.596
implicit gradient	4.56913×10^{-8}	1.42457×10^{-7}	1.50013×10^{-7}	n/a	0.254

Table 10: Three learning rules on one mechanism, three teachers, 256 test cases per condition. The rule that reads two endpoints learns what the tape learns, to four digits, from an eighth of the retained state. The gradient controls fit faster in these implementations.

This is evidence for conditional numerical reuse on a large sparse graph. Establishing the same benefit in an adaptive agent requires counting initialization, memory, changed-source discovery and scheduling over its actual stream.

7.5 Credit without a tape

A symmetric 48-neuron net with unknown weights, 16 driven inputs, 16 observed outputs, three teachers, output targets only. The centred contrast, backpropagation through 64 tied settling iterations, and an implicit gradient receive identical initial weights, minibatches, Adam arithmetic and stability projection [R13].

That row is Lemma 4.7 and Fact 4.8 on a real net. The consequence is not accuracy. It is that the update can be computed at the synapse while the brain is running, which is what the lives of Table 6 do and what a frozen stack does not.

7.6 Damage, and a body that changes

One pass of 30,000 images; then half the pixels dark for good, or half the hidden neurons removed; then a hundred updates of continued learning under each learner’s own rule [R16]. The patch net goes 0.912 intact, 0.822 blind, 0.899 recovered, and 0.794 lesioned, 0.896 recovered: 0.85 of the loss regained after blindness where Adam regains 0.50, and 0.87 after the lesion where Adam regains 0.86. The patch net loses more at the moment of damage. Theorem 4.5 bounds the displacement of the equilibrium; recovery is the measured behaviour of the same local rule that learned.

The arm’s lower link is lengthened during the same life: target success is 0.992 with the changed body and 0.991 after restoration [R51]. The artist reaches foreground scores of 0.926 and 0.927 under the corresponding changes [R54]. Connect Four’s score against earlier opponents changes by +0.001 after 200 games against a new mixture [R53]. These are bounded tests of adaptation and retention; no theorem guarantees that all previously correct records survive a new body or task.

7.7 Planning in the head

A patch net can imagine because it has a model it owns: the records answer what would follow a reading, and the settling composes the imagined reading. Connect Four is the clearest case, because the quality of a move can be judged exactly.

	one-ply	minimax	solver 50%	solver 70%	AlphaZero	trap suite
the brain, 1,200 games of one life	0.985	0.805	0.870	0.630		
the brain, deeper search	1.000	0.945	0.935	0.745	0.945 / 0.990	0.895
the brain, threats read off the records	1.000			0.950		0.915
AlphaZero, published recipe	0.738	0.688				
minimax, 1,024 nodes	0.875		0.670	0.395	0.375	0.570
one-ply	0.585		0.585	0.200		0.505

Table 11: Connect Four against graded opponents, paired games with the candidate opening the even games [R56,R60]. The minimax column is the supplied-rules opponent at 1,024 nodes against a matched budget of 1,024 imagined transitions. AlphaZero is the published alpha-zero-general recipe at 24 iterations, met at 25 and at 100 simulations per move over 100 games; that opponent scores 0.738 against one-ply and beats minimax 0.625 to 0.375. The trap suite is 200 positions with a forced line four to ten plies deep, scored by a perfect solver. The threat row is the same frozen records with the threat read switched on, 40 games. Against a perfect opponent moving first every arm scores 0.0: the first player wins Connect Four with perfect play.

The brain sees a board as categorical cells and returns a column. It is given no board object, no rules engine, no terminal oracle and no correct-move labels. It learns from its own games what a dropped stone does, which windows of four cells are completed lines, and, in the current life, which positions its own searches proved and which columns the winners of the games it watched played. It searches over boards it composes itself with alpha-beta pruning inside a budget of imagined transitions, and reads the threats standing on a board off the same line records; the rule that turns threats into a value is supplied, like the search [R53]. After 1,200 games of one life, the exact next board is right on 1.00 of 10,000 held-out moves, terminal prediction is 1.00 and the tactical suite is 1.00 on every seed.

Table 11 evaluates a learned transition model inside supplied alpha-beta search and a supplied threat heuristic. Reading threats improves the score against the 70 percent solver from 0.825 to 0.950 and the trap suite from 0.895 to 0.915 without further learning. These bounded opponents and search budgets do not establish superiority over AlphaZero as a method. The artist offers a more direct feedback test: after a canvas offset its foreground score is 0.708 with readback and 0.436 without it [R54]. Its planner can revise a stroke because the next observation reports what the previous action actually did.

Planning makes a learned consequence model useful for action. The supplied search, candidate budget and heuristic are part of the result and its cost; readback then tests the chosen action against the world. This loop, rather than any single game win, is the capability relevant to the hypothesis.

7.8 A life that does not freeze

Thirty permuted-MNIST tasks, 20,000 images each, seen once, one head, no replay [R17]. The patch net learns the first task at 0.924 against Adam’s 0.920, averages 0.913 against 0.921, and ends the thirtieth at 0.894 against 0.918; a patch transformer averages 0.840 and plain stochastic gradient descent 0.837 [18]. Every learner has forgotten the first task by the end, and the patch net drifts three points where Adam drifts none. That is a loss, recorded in Section 10. Inside a life the same property reads differently: the worm connectome of Section 9 relearns a reversed odour association to 0.96 or better in every seed [R21], the arm absorbs a changed body and returns [R51], the Connect Four brain keeps its old opponents while learning new ones [R53], and a brain that watched nine recordings of one console game and then learned from its own play improves from 626 to 720 pixels on the trained level and from 372 to 806 on a level it never watched [R26].

8 Where the wiring comes from

The application layouts in Section 7 are hand-wired. In the composer, splitting a joined reading into pitch and gate heads was consequential: neighbouring pitch classes shared 0.567 of their active cells at the original offsets and 0.010 at zero [R65]. Automating such representational choices is a central research question. The following experiments test search over recorded streams, allocation during life and ecological selection, each within a supplied mechanism vocabulary.

family	logged streams		intervention fitness		winner's wiring	
	evolution	random	evolution	random	accuracy	floor
distractors: one relevant field among eight	+0.540	+0.228	+1.705	+0.386	0.660	0.75
conjunction: two fields jointly	+0.737	+0.380	+1.260	+0.952	0.896	0.80
context: the rule depends on a context unit	+0.291	+0.056	+0.514	+0.164	0.508	0.55
effects: the action decides the outcome	+0.939	+0.856	+1.972	+1.671	1.000	0.95

Table 12: Wiring search on four synthetic task families with a labelled oracle mask, 432 measurements per arm, skill over the joined head on the untouched part of the stream [R61]. Evolution is ahead of the random search on every family under both fitnesses. By accuracy on the logged streams the winner clears the floor on two families; under the intervention fitness the distractor winner predicts the effect of an action at a factual skill of 0.921 and is admitted. On the context family the winner drops the context unit and fails, which is the family where the oracle mask’s own ceiling is 0.560.

8.1 Search over the wiring on a frozen stream

The genome of one record head is a reading mask over the stream’s fields, per-field gains and offsets, the number of cells and winners, a fan-in, and a target field with a delay. The fitness is read from a recorded stream and never from a life: the head’s held-out prediction skill on the required fields, less a charge for cells and synapses, and, under the intervention fitness, the head’s skill at predicting the effect of an action actually executed from a saved state. Two searches run at equal budget, twelve generations of twelve, 432 measurements each: an evolutionary loop with elites and mutation, and a random search over the same prior. The wiring a person found by hand is measured in the same receipt as a disclosed ceiling, and a joined head reading every field is the start every stage writes first.

The composer test carries a substantial prior: intervened target heads must read the swept action field, here pitch. The search can remove other inputs and tune the head; it does not discover the relevance of pitch from an unnamed slot. At a two-block horizon, with 3,510 chip blocks charged for intervention states, the selected evolutionary winner scores +0.368 against random search’s +0.063, the hand-built head’s +0.086 and the joined head’s −0.436 [R61]. Its pitch-only reading uses offsets 0.02, 1,296 cells and 55 winners, found in 251 seconds against 806 for random search. Untouched-stream scores of +0.320 and fresh-stream +0.102 are positive, but both fail the all-field requirement because the sound-presence prediction fails. On a second stream recorded per held note, random search wins, +0.853 against evolution’s +0.438. This is evidence for useful restricted automation, not a general efficiency advantage for evolution.

Transferred into a life, the evolved head improves early pitch accuracy: across four seeds it reaches 0.384 after 6,160 blocks, against 0.103 for the hand-built head. The latter reaches 0.395 at 12,342 blocks and random search’s head reaches 0.397 at 24,926; all end near 0.52 at 50,162 blocks [R65]. The result is faster early acquisition at these checkpoints, not twice the eventual competence. On the 1942 stream, 400 intervention states with eighteen actions each yield no selected wiring better than predicting no effect; the evolutionary winner omits the action and scores 0.000 [R61]. That negative result limits this representation and horizon.

8.2 Allocation during a life, and the budget curves

The allocator starts with a bank reading the full stream. Residual-weighted statistics propose masks from a supplied vocabulary of single coordinates, pairs and the full reading; candidates face probation, with at most six banks and a context router. The policy and mask vocabulary are hand-specified; the selected coordinates are learned. Fourteen arms run on four families at 5,000, 20,000, 80,000 and 320,000 decisions, with ten, ten, three and two lives per family and preregistered floors [R62]. Guards that protect records across world changes are evaluated separately.

The allocator reaches 0.946 mean accuracy after 5,000 decisions and clears every floor on all forty lives, against 0.490 for the online transformer and 0.341 for the gradient net. At 320,000 decisions the transformer reaches 0.822. This is a sample-efficiency comparison, not matched memory: at 5,000 decisions allocator state occupies 1.63–13.33 MB across families, against 0.087–0.112 MB for the online transformer. The fixed records learner plateaus after 20,000 decisions, while local contrast alone falls from 0.242 to 0.202 as the budget grows. More experience therefore does not automatically improve the slow learner.

Acquisition does not imply retention. After learning a second world, the plain allocator recalls the first

arm	at 5,000	at 320,000	floors cleared at 320,000	lives clearing
in-life allocation, context router	0.946	0.981	4 of 4	8 of 8
in-life allocation, uniform mixture	0.873	0.969	4 of 4	8 of 8
gradient net, encoder pinned at half budget, ring	0.336	0.866	4 of 4	8 of 8
the oracle’s mask (a labelled ceiling)	0.796	0.864	4 of 4	8 of 8
online transformer, ring	0.663	0.836	3 of 4	6 of 8
online gradient net, ring	0.541	0.825	3 of 4	7 of 8
online transformer	0.490	0.822	3 of 4	6 of 8
online gradient net	0.341	0.769	3 of 4	6 of 8
the fixed records brain	0.518	0.551	1 of 4	2 of 8
local contrast, settled regions	0.242	0.202	0 of 4	0 of 8

Table 13: Mean accuracy over the four task families against the budget of decisions in one life [R62]. In-life allocation clears every floor on all forty lives at 5,000 decisions; the online transformer needs sixty-four times the budget to clear three of four and stays at chance on the conjunction family at every budget. The fixed records code is flat after 20,000 decisions and the local-contrast arm falls with budget.

at 0.599 at the 5,000-decision budget and 0.261 at 320,000. A combined learner adds a conventional gradient patch, a counted replay ring and protected banks. On the 320,000-decision distractor task, the guarded and unguarded versions both acquire at 0.967, while old-world readback is 0.945 with the guard and 0.163 without it, two lives each [R63]. This result belongs to the combination, not local contrast alone. Calibration of a broader developmental policy reaches 0.961, 0.978, 0.945 and 1.000 on four development families but only 0.418 and 0.575 on the delayed-cue and returning-regime transfers [R64]. It does not establish general transfer of a learned developmental rule.

These experiments support automatic selection of useful input views and record allocation within a supplied grammar. They separate acquisition from retention and expose the price of the memory. They provide no example of a mechanism invented outside the genome or allocator vocabulary.

8.3 Ecological selection of modular wiring

Patch World replaces an explicit fitness ranking with reproduction and death in a supplied ecology. A 96×96 torus conserves mass across soil, food and creatures, with drifting light and local interaction rules. A genome specifies up to four record cortices, masks over fourteen sensory groups, sizes, active counts, fan-in, sensing radius, planning horizon and learning rates. Reading, storage and writes cost mass. Energy thresholds govern reproduction; children inherit mutated genomes and empty records. The learner’s internal reward is mass change. Mutation operators explicitly include adding, removing and cost-neutrally splitting a cortex. Thus evolution selects combinations of available parts; neither the parts nor the splitting operation emerge. Runs start with 300 founders and span 6,000–8,000 ticks on two seeds [R67].

The clearest architectural response occurs in rock-and-bite worlds when reading costs a quarter of the base price. The final populations average 3.08 and 2.14 cortices per creature; one contains 167 three-cortex and 126 four-cortex creatures among 361. One-step planning occurs in 201 of 361 and 324 of 513 creatures, with longer horizons also present. Under the original read price, biting worlds often lose food sensing instead. This contrast supports selection shaped by the cost of information; two seeds and descriptive population comparisons do not isolate each mechanism’s causal contribution.

The receipt’s “halves” statistic means at least two cortices plus a reading mask that includes the prior-code channel. It is 8 and 10 percent in those two cheap-reading worlds and 28 and 15 percent with free reading. The channel makes previous code summaries available for sampling, including a cortex’s own code and unused slots. The statistic checks neither bilateral symmetry nor actual cross-cortex synapses, reciprocity or functional dependence. Median lifetimes of marked versus unmarked creatures are 37 versus 33 ticks in one free-reading seed and 30 versus 31 in the other. There is no consistent longevity advantage. Hearing is associated with reward of 0.50 versus 0.44 per tick, without a causal intervention; symbols carry only 0.001–0.004 bits about the measured food, neighbour and kin variables.

Patch World provides exploratory evidence that ecology and sensing costs can select multi-module record brains from a supplied grammar. It does not demonstrate evolved bilateral organization or a hierarchy of mutually necessary modules. Establishing functional internal readback requires checking the sampled connections and intervening on them; module count alone cannot supply that evidence.

9 A worm that reacts and learns

The *C. elegans* experiment makes the computation visible at the level of named cells. The fixture derived from the hermaphrodite connectome [14] contains 300 connected neurons. It lists 3,638 chemical connection pairs and 1,093 gap-junction pairs. Removing 47 self-pairs, making gap messages bidirectional and merging parallel messages gives 5,044 directed computational links. CANL and CANR have no synaptic edges in this fixture and are omitted. Measured wiring determines the graph; Cadence supplies the neuron equations and fitted parameters. Gap-derived links carry weighted activations, without an electrical conductance model.

9.1 A collision, seen from inside the network

Figure 5 shows the entire graph inside a schematic worm outline and follows one actual boundary encounter from the trained browser model [R70]. The model body is a point controller in a circular dish: it receives odour comparisons and touch, chooses a turn or forward action, and reads back the result. The drawing exposes its internal computation; it does not simulate muscles or the deformation of a living worm.

1. Contact changes the sensory drive. An attempted movement crosses the dish rim. The body code blocks that movement and activates the six supplied FLP/OLQ touch inputs. Their held potentials describe the preceding reading. The changed drive therefore produces a local equation residual, $r_i = \sum_j W_{ij} \text{act}(v_j) + c_i - v_i$.

2. Local updates spread the disturbance. Each neuron moves its potential toward the value its inputs require. Its changed activation alters neighbouring inboxes, so a disturbance initially concentrated near the input can recruit distant cells through recurrent connections. There is no prescribed sequence of layers. In this trace, nine neurons have absolute residual above 0.01 at the changed input, 183 do after four updates, and none do after 35. The solver uses synchronous sweeps; locality describes the dependencies, not a measured saving from skipping neurons.

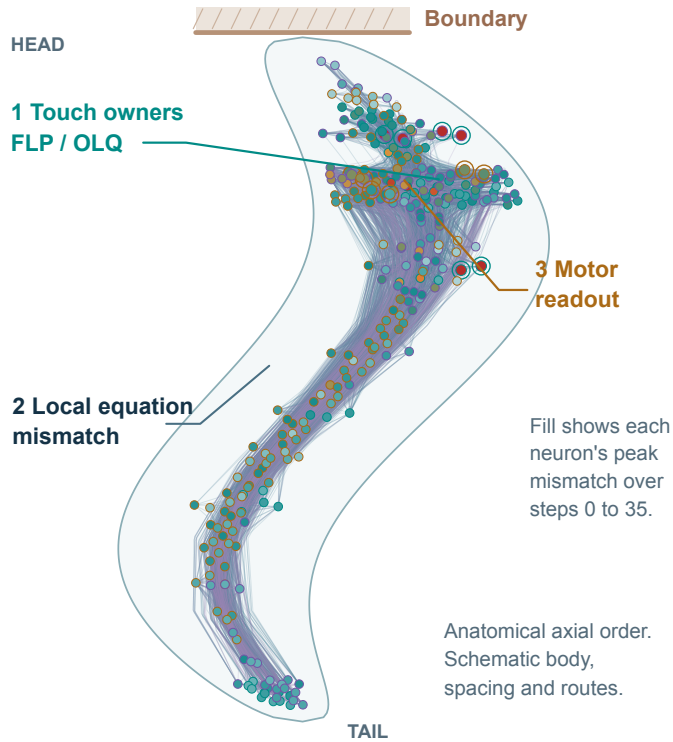
3. The motor readout changes the next action. The supplied output pairs SMDD, SMDV and AVB control left, right and forward choices. With identical post-contact odours, switching touch off gives a right-turn probability of 32.2%; with touch present it is 99.4%. The continued rollout clears the blocked boundary after five actions. This is one controlled illustration of a response, not an obstacle-avoidance benchmark. The trained parameters are frozen throughout the encounter: settling changes neural state, while learning changes the rule governing subsequent encounters.

A boundary encounter through the whole worm graph

Measured *C. elegans* wiring; a recorded response in the Cadence model

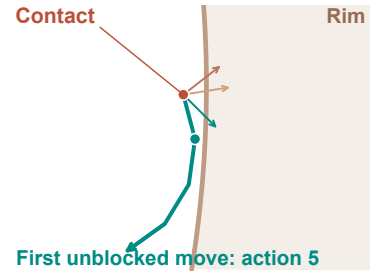
300 modeled neurons / 3,638 chemical pairs / 1,093 gap pairs

A Touch input, local repair, motor readout



B The recorded motion

Point-body trajectory at the rim



C Touch changes the policy

Right-turn probability

Same odors and starting state

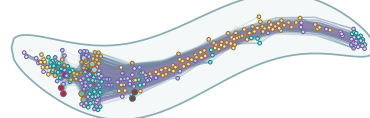
Touch off 32.2%

Touch on 99.4%

Frozen weights. One encounter. No new learning in this trace.

D Three actual settling states

Step 0



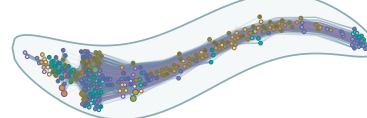
Max mismatch 3.0006

9 neurons above 0.01

Fill: absolute equation residual



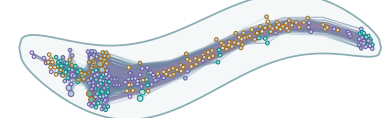
Step 4



Max mismatch 0.3605

183 neurons above 0.01

Step 35



Max mismatch 0.0041

0 neurons above 0.01

Outline: sensory

interneuron

motor

Figure 5: **A disturbance becomes an action on the complete connected worm graph.** All 300 modeled neurons and all source connection pairs are drawn; execution uses the 5,044 directed links described in the text. Soma order follows anatomical coordinates, while spacing, body shape and edge routes are schematic. Colours show computed local equation mismatch, with a common scale across the recorded stages. The body trajectory and motor probabilities come from the frozen model, including a control with identical odours and touch removed. Solver steps are not biological time. This figure depicts inference; Table 14 tests learning and reversal.

9.2 What experience changes

In the learning experiment, food is associated with one of two odours, read through AWA and AWC. The agent samples actions, observes outcomes and uses a critic’s reward error to modulate local eligibility traces on 941 designated links; neural biases are frozen. Free and perturbed settling states supply endpoint contrasts. Reward combines food discovery with a supplied approach signal using the simulator’s food location. Thus sensory mismatch drives the state toward a response, while the engineered reward determines which action-related changes to reinforce. The measured graph is directed and asymmetric; applying this learning rule does not establish the symmetric-network gradient identity for the worm.

The three-seed comparison uses sixteen parallel bodies sharing one set of parameters, trained for 8,000 batch updates, or 128,000 body transitions per seed [R21]. With learning disabled at evaluation, the measured wiring finds food in 99.9–100.0% of fresh searches, against 21.3–22.9% before training. Shuffling the source connections yields 52.6–87.7% after the same training budget. The shuffle preserves the raw contact multiset, but merging produces different numbers of computational links. A separate single-body run reaches 100% on 500 evaluation searches after 48,000 decisions; the main three-seed result should therefore not be read as a single 8,000-transition life.

Cells clamped off	Diacetyl rewarded	Butanone rewarded after reversal
None	99.9–100.0%	96.2–99.5%
AWA (diacetyl input)	25.0–35.8%	94.4–99.1%
AWC (butanone input)	99.7–99.8%	24.3–29.4%
RIA (interneurons)	18.9–21.8%	15.9–26.1%

Table 14: **Learning changes which named cells the policy needs.** Food-finding success, range across three seeds, with at least 1,000 fresh searches per seed and intervention, sampled actions and learning disabled. Reversal continues from the acquired parameters for another 8,000 batch updates with the odour–food association exchanged [R21]. The first column comes from acquisition runs; the second from their continued reversal runs. Interventions clamp named cells in the model, including supplied sensory inputs and internal RIA interneurons.

After reversal, AWC becomes necessary for high success and AWA loses most of its influence (Table 14); RIA matters under both associations. This change in intervention dependence connects behavioural learning to named parts of the graph. It provides a more informative mechanism test than activity on the connectome alone. A smaller multilayer actor–critic learns the task faster, and connectome-constrained behavioural models have precedents [34]. The contribution here is the explicit relation between local computation, experience and interventions, without a claim that these equations reproduce the living worm’s physiology. The archived learning-worm evidence exposes the circuit and action loop.

9.3 Other measured connectomes

The adult fly connectome at release 783, 138,639 neurons and 2,700,513 directed synapses [19, 72, 74], settles as one net. Thirty single-channel exposures and 28 held-out mixtures, with localised and distributed lesions of 1,024 and 2,048 neurons, are answered with global residuals below $9.8 \cdot 10^{-9}$ and a uniform potential-error bound of $6.7 \cdot 10^{-8}$, and the library agrees with an independent cold solver to $6.1 \cdot 10^{-16}$ [R22]. Under the published model’s drive convention the untrained wiring passes 14.75 of 29 held-out experimental facts against 7.5 for a shuffled wiring [R23]. The regionalised mesoscale mouse connectome [37], 359 leaf regions per hemisphere and 61,454 projections, is given whiskers, a retina, a nose, a vestibular sense and a place code, with only the three-neuron cord’s 6,462 synapses plastic [R24]: in a labyrinth with food under one of two odours the measured wiring reaches 0.953 held-out success on 64 fresh labyrinths at 1.56 times the shortest path, the shuffled wiring 0.609 at 3.59 times; one seed each, and a torch actor-critic reaches 1.000 in 12 seconds against 46 minutes.

These connectome rows show that the software can instantiate supplied measured connectivity and that this connectivity can matter under the chosen dynamics and sensory interface. They do not establish that those simplified dynamics reproduce a nervous system’s computation. Conventional solvers and other neural implementations can also use the same graph.

10 Drawbacks

The revised core adds an explicit interaction and evidence contract to the existing recurrence; it cannot claim the performance of a larger application that uses additional mechanisms. Its constructor checks reciprocal structure, and its learner checks phase residuals. These checks do not establish global energy descent, multiple useful memory basins, long-delay credit, autonomous role development or resilience to ongoing incompatible learning. The following measured limitations concern the named earlier implementations and controls.

1. **The layout is designed by hand.** Every brain of Section 7 and Appendix D was wired by a person, and the wiring was where the time went: the composer’s factorisation cost six rounds and thirty-nine full lives [R65]. Search and evolution have replaced the hand on one record head, on synthetic families, and in a world of small brains; the allocator’s policy parameters are set by hand; the genome configures a mechanism and does not invent one; and the player’s stream defeats the whole search family [R61,R62,R64].
2. **The slow learner.** The settled cortices trained by the local contrast alone fall with budget on the task families, 0.242 to 0.202, and unlearn inside a long life [R62]; the largest brain loses to a recurrent unit on its own corpus, 1.023 against 0.708 [R68]; the composer’s settled regions decide nothing [R65]. The records are a fast memory; the pure local-contrast learner does not turn larger budgets into competence on these families.
3. **Wall-clock.** In the arm’s own life a decision costs 12.9 ms at the 95th percentile against 1.0 ms for the multilayer network [R51,R55]; on supervised rows the factor is 13 to about a hundred [R30,R17,R16,R5]. The player’s conventional arm is 9 to 35 times cheaper per decision [R66], and on the composer the accelerator path is slower than the CPU path [R65].
4. **Language.** On character-level text the patch net is behind at every width tried: 3.29 bits per character at width 128 against 3.22 for a same-window multilayer network and 3.03 for a one-layer transformer whose validation reaches 2.67 [R31,R30]. The joint settling of an utterance is the thesis under test and it is not confirmed [R32,R45].
5. **Hard temporal credit and composed futures.** From reward alone the three-factor rule reaches 174 on Hopper at 300,000 steps against about 500 for proximal policy optimisation, 0.64 to 0.73 of balls on Pong against 0.98, and 488 on a slippery ant at two million steps against 1,373 [R33,R34]. Correcting the policy nudge raises a delayed binary choice from 0.50 to 0.88, where a ten-parameter conventional score rule reaches 0.98 [R44]. Neither reported arcade controller clears a stage [R66,R69].
6. **Retention across worlds.** On split MNIST with one head every learner, this one included, ends between 0.18 and 0.20; a reservoir of 256 stored rows raises it to 0.78 [R36,R42]. On thirty permuted tasks the drift is three points where Adam drifts none [R17,R46]. Read back after a world change, every learning arm of the budget curves loses the old world, and the guard that repairs it for the record banks fails on the effects family above 20,000 decisions [R62,R63].
7. **Supplied addresses, supplied searches, supplied rules.** The position tags of the span task, the clock of n -back, the place keys of the world and the receptive fields of the Connect Four cortices are supplied, and so is every search and the rule that turns threats into a value [R43,R53]. What the brain learns is the model those searches consult.
8. **Conventional controls that win.** A dictionary is exact on every memory row. A windowed transformer is perfect on n -back. A sparse solver settles the fly a tenth faster than the library. A supplied 32-step recurrence answers worm lesions to 0.03 percent [R37]. A multilayer network with a replay ring reaches 0.980 against 0.994 in the arm’s life at a twelfth of the latency [R55], and a gradient net with a pinned encoder and a ring clears every floor of the task families at sixty-four times the budget [R62].
9. **Conditions of the identity and of the certificate.** Asymmetry among free neurons, adapting state and unconverged phases need separate treatment, and finite nudges are biased; a five-seed derivative control measures the residual bias [R41]. Theorem 4.1 is stated without adaptation, and the trained nets of several rows were not checked against its row-mass condition.

10. **Seeds and scale.** The four lives use five acceptance seeds each; the budget curves use two lives at the top budget; the evolution world uses two seeds; historical rows use one to three. The largest brain here holds seventy million synapses, four orders of magnitude below a frontier language model, and nothing in this paper establishes that the design scales to that size.

11 Open challenges

The hypothesis of Section 1.3 depends on several capabilities that the following tests distinguish.

1. **A layout that generalizes beyond its grammar.** Restricted search improves one pitch head, allocation finds predictive coordinates, and ecology selects module counts. Stronger evidence requires learned developmental rules that transfer to new task structures, plus causal tests of the resulting modules. The calibrated policy’s weak delayed-cue and returning-regime transfer [R64] and the player’s zero intervention skill delimit the demonstrated result.
2. **Useful learning through the common primitive.** The earlier records write a fact in one shot and plateau when their code is used up; the tested local-contrast cortices do not improve reliably with budget. A separate gradient patch improves from 0.359 to 0.872 in the combined learner [R63]. This is a diagnostic control. The architectural hypothesis requires useful continuing acquisition through repair and detuning, with warm execution, teaching signals, parameter constraints and numerical convergence tested together.
3. **A learned organization that decides.** The hand-wired world and composer brains include workspace regions, and in the composer the planner reads the moment and the records while the workspace takes no part [R65]. A readable trace can retain a cue while contributing too little to action [R81]. Useful specialization requires causal interventions on acquired roles, beyond unequal weights or clusters. The settled network must contribute to action and evidence-seeking, while memory is accounted for in its actual persistent state. Neither a named workspace nor a generic counterfactual branch establishes those functions.
4. **Retention, revision and forgetting together.** Sparse writes protect what a reading does not touch; a new world’s readings can touch old records, and the tested guard fails where two banks fit two worlds equally [R62,R63]. The nonlinear diagnostics add two different failures: stable activity can block legitimate revision, and stable individual contacts can exclude useful continuous relationships [R72,R73]. Supplied selective plasticity protects four associations through learning and correction [R78]; learning that separation while retaining useful shared structure is a different requirement. Learned importance must improve this joint task rather than merely preserve stronger supplied signals. Finite precision and capacity, false strong observations and later conflicting learning must all be included.
5. **Credit across many decisions and composed futures.** One-step consequences are learned well everywhere; a future composed from them is not trusted by the policy that reads it, and no player clears a stage. The three-factor rule is behind the conventional score rule at long delays [R44,R66].
6. **Bodies and real time.** A robot that learns on the job needs the arm’s property, a body absorbed and returned without a reset, at the latency of its actuators. The arm decides in 12.9 ms and the player in 46 ms at its size and 201 ms at four times its size [R51,R66]; the certificate bounds the error of a decision returned early, and the latency gain from a different substrate must be measured rather than inferred from locality.
7. **Scale and the sequence learner.** On text and on the corpus of the largest brain the conventional sequence learner wins. The hypothesis does not need a patch net to be the better corpus model, but it does need the mechanism behind the composer’s loss understood, and the second item is the current answer.

12 Conclusion

Cadence’s revised architectural hypothesis is that local nonlinear repair and equilibrium detuning can support a continuing individual whose activity interprets the present and whose learned relationships preserve useful experience. The library makes those operations explicit: persistent free state, witnessed port losses, local parameter contrasts, numerical phase checks and isolated imagined branches. These are implementable contracts, without a claim that the network has acquired general memory or a biological mind.

The mathematical boundaries identify what the primitive needs. Global contraction erases fixed-input activity history. A sufficient memory must distinguish histories with different required futures. A causal poset can organize evidence without supplying its missing content or semantics. Basin retention, scalar trace repair and selective contact decay hold under explicit, limited hypotheses. Supplied noninterference permits successful parameter retention and selective correction in a small network, at increased integration cost; a fading internal intention can nevertheless fail to guide action. These results distinguish retaining a signal, learning where parameters may change, and using the retained signal in the whole actor.

The earlier application evidence supports learning from consequences with additional records and supplied organization, and the certified numerical subclass supports conditional savings from state reuse. Their successful scores are evidence for those combinations. Testing the revised hypothesis requires useful acquisition, retention, correction, specialization and active evidence collection through the common patch operations, followed by transfer across bodies and tasks at competitive total cost. The identity between an additive record and a linear-attention read proves a property of that record; it does not establish that attention is universally unnecessary.

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A Theorem map and the Lean library

Every Theorem, Lemma and Corollary in the paper names the Lean declaration that proves it. The public Cadence library includes the certified recurrence, activation, local contrast and wiring results, together with the representation, noise, basin, specialization and temporal-overlap boundaries [53]. The optional linear-record algebra has a separate import; it is not a component of **PatchNet**. These sources pin Lean 4.29.1 and Mathlib at the matching release. Their axiom audit allows only propositional extensionality, choice and quotient soundness, with no admissions or project-specific axioms. The historical cost results are retained in the full manuscript’s proof collection; exploratory structural-memory and six readable-trace results have separate sources and verification in the accompanying memory evidence [54]. The trace lemmas are additional to the public package’s 156 audited statements. The full manuscript build checks these additional sources as well. None of these collections is a formal refinement proof of the floating-point experiments or the complete library interface.

Table 15: The theorem map. Settling theorems hold for Lipschitz activations; the Activation module instantiates them for this neuron. Memory identities hold over commutative rings, the capacity bound over fields, the cost statements over the naturals.

paper	module and declaration	content
Thm 4.1	Settling, <code>settle_contracting</code>	contraction with constant $1 - dt(1 - L\rho_W)$
Cor 4.2	Settling, <code>equilibrium_unique</code> , <code>silence</code>	any fixed point is the equilibrium; rest is the equilibrium of no input
Thm 4.3	Settling, <code>apriori_bound</code> , <code>aposteriori_bound</code>	the two error certificates
Thm 4.4	Settling, <code>equilibrium_lipschitz_stimulus</code> , <code>warm_start_bound</code>	the equilibrium moves by at most $dt\delta/(1-q)$; k warm steps leave $dt\delta q^k/(1-q)$
Thm 4.5	Settling, <code>lesion_bound</code> , <code>lesion_contracting</code>	a lesion moves the equilibrium by at most $dt\mu_S A/(1-q)$
Thm 4.6	Activation, <code>activation_lipschitz</code> , <code>activation_abs_le</code> , <code>activation_zero</code> , <code>learningActivation_lipschitz</code> , <code>learning_model_certified</code>	the neuron’s Lipschitz constant, range and rest; row mass below two certifies the learning model
Lem 4.7	Credit, <code>contrast_local</code> , <code>contrast_symm</code> , <code>contrast_eq_zero_of_agree</code>	locality, symmetry, zero contrast for identical states
Thm 4.9	Memory, <code>read_write</code> , <code>read_write_self</code> , <code>read_write_orth</code> , <code>repeat_write_stable</code>	interference law, one-shot write, orthogonal invariance, idempotence
Thm 4.10	Sparse, <code>read_write_dist</code> , <code>read_write_dist_le</code>	the exact size of one write’s effect on another read

paper	module and declaration	content
Thm 4.11	Memory, <code>read_writeUpTo</code> , <code>capacity_le</code>	exact storage of an orthonormal family; at most d such keys
Thm 4.12	Sparse, <code>dotProduct_eq_zero_of_disjoint_support</code> , <code>read_write_of_disjoint</code> , <code>read_writeUpTo_of_disjoint</code> , <code>disjoint_family_card_le</code>	disjoint supports never interfere; a disjoint family is stored exactly; a code of width D with k winners holds at most $\lfloor D/k \rfloor$ of them
Thm 4.13	Memory, <code>read_hebb</code> , <code>one_hot_recall</code>	the additive read is linear attention; one-hot recall is exact
Thm 4.14	Wiring, <code>rowMassLE_of_abs_le</code> , <code>rowMassLE_smul</code> , <code>rowMassLE_add</code> , <code>mutation_contracting</code> , <code>growth_contracting</code> , <code>population_contracting</code>	pruning, masking and rescaling keep the certificate with the same constant; growth keeps it inside the row-mass budget; one bound certifies a whole population
Thm 4.15	Cost, <code>attention_total</code> , <code>attention_total_ge_quadratic</code> , <code>record_total</code> , <code>record_state_const</code> , <code>exists_crossover</code>	the transformer’s arithmetic over a stream is quadratic in its length and its cache unbounded; the patch net’s is linear with constant state; a crossover exists

Three further results are used as facts and are Lean-checked in the observer-patch corpus rather than in this package [56, 58]: local repair to consensus on a federation of patches computes every finite Boolean function (`echosahedral_universality`); a work-conserving scheduler on a ranked repair system terminates without a fairness premise; and the accepted repair steps of a fixed input-independent federation number at most $n(n+1)/2$, order-sharp. No embedding of those discrete programs into this neuron model is supplied.

B Evidence map

Empirical numbers are read from the named receipts copied into the evidence package, with each receipt’s own digest where it has one, its implementation version or source hashes, and the sha256 of the copied bytes recorded in the package manifest. Quoted external results have bibliographic citations. Exact mathematical constants follow their stated definitions and hypotheses. The memory-proof sources are copied under the manifest label `mission-memory-proofs`.

Table 16: Receipt identifiers and manifest labels. The library column names the release whose sources the receipt binds; OPN identifies the earlier observer-patch-nets library. The manifest preserves exact source references and copied-file hashes.

id	manifest label	receipt	library
R1	recall	associative recall over 128 pairs, two transformer arms	0.5.0
R2	revision-memory	memory under a revision stream, five arms, three seeds	0.8.1
R3	pattern-separation	correlated keys under revision streams, six arms, five seeds	0.9.0
R5	digit-span	digit span, six arms, fade sweep	0.7.0
R6	n-back	n -back, four arms, two baselines	0.7.0
R8	adaptive-depth	compute follows change: warm steps, budget, bounds, three row masses, five seeds	0.9.0
R10	event-repair	event-driven repair on the fly graph, 135 conditions	0.8.1
R12	surprise	settling steps against entropy	0.5.0
R13	matched-learning	three learning rules on one 48-neuron mechanism	0.8.1
R16	lesion	damage and recovery, three seeds	0.5.0
R17	permuted-mnist	thirty permuted tasks, six arms	0.5.0
R21	worm-learns	the worm learns an odour, three seeds, lesions, swap	0.8.1
R22	fly-counterfactuals	whole-fly counterfactual screening	0.8.1
R23	opn	fly wiring against shuffled, conformance	OPN
R24	mouse	mouse labyrinth, connectome against shuffled and a network	0.7
R26	player	self-play of the many-games brain	0.7
R27	player	sixteen games at once with a dopamine floor	0.8
R30	ladder-v0.5.0	digits, MNIST, Connect Four, Pong, text, sign, chorales, cart-pole, worm	0.4

id	manifest label	receipt	library
R31	sizing	text width sweep, per-run files	0.4 to 0.7
R32	author	language pilots at level one	0.7
R33	hopper, pong	control gates, logs	0.5.0
R34	repeat-previous, slippery-ant	partially observed cards; slippery ant	0.7 to 0.8.1
R36	split-mnist	split MNIST, one head	0.5.0
R37	worm-probe	supplied worm mechanism under lesions	0.8.1
R41	ep-inputs	fixed-input derivatives and free/free asymmetry control	0.9.0
R42	rehearsal	split and permuted MNIST at disclosed memory and work budgets	0.9.0
R43	content-prototypes	noisy cues, fixed prototypes, aliases and capacity	0.9.0
R44	policy-credit	delayed choices, loss-policy correction and score control	0.9.0
R45	sequence-readback	causal held-out text and validation grids	0.9.0
R46	drift-diagnostic	thirty tasks and equal-budget reset forks	0.9.0
R51	experience-arm	the arm: five acceptance seeds, twelve gates, six controls	0.9.0
R52	experience-world	the world: five acceptance seeds, eleven gates, eight controls	0.9.0
R53	experience-connect-four	Connect Four: five acceptance seeds, twelve gates, the threat read and the memory of proved positions	0.9.0
R54	experience-artist	the artist: five acceptance seeds, thirteen gates, eight controls	0.9.0
R55	experience-comparisons	the same lives with the world model replaced by a multilayer network or a transformer, with and without a replay ring	0.9.0
R56	solver-bench	Connect Four against graded solver opponents, every move scored by a perfect solver	0.9.0
R57	instrument	the instrument: five acceptance seeds, ten gates	0.9.0
R58	composer-stage	the chip composer: five acceptance seeds, thirteen gated and seven open measures	0.9.0
R60	connect-four-bench	Connect Four against a published AlphaZero recipe, the threat-read ablation, the trap suite	0.9.0
R61	wiring-search	wiring search on four families and on the composer's streams: evolution, random search, hand-found wirings, intervention states	0.9.0
R62	scaling-curves	fourteen arms on four task families at four budgets, preregistered floors, retention read-back	0.9.0
R63	combined-learner	banks beside a gradient patch with a guard, ten arms, four budgets	0.9.0
R64	evolved-development	the evolution loop over wiring and development policy: contract and calibration	0.9.0
R65	composer-curves	the chip composer as curves over a life: three wirings, two sizes, two life lengths, four seeds	0.9.0
R66	player-curves	the 1942 records controller and a PPO convolutional net at matched frames, three seeds	0.9.0
R67	world-evolution	the evolution world: five sweeps over rules and prices, two seeds each, censuses and lifetimes	0.9.0
R68	maestro	the 68-million-synapse musician: receipt, model card and evidence sheet with the recurrent control	0.9.0
R69	player-1943	the 1943 imitation player and its practice on 1942, frozen tests on eight starts	0.9.0
R70	worm-contact	pinned browser sources, complete boundary-contact trace and matched touch-off control	0.8.1
R71	mission-basin	nonlinear cue persistence, opposing evidence and continued-learning stress; source, full receipt and independent verification	OPN
R72	mission-specialization	nine-patch completion and warm revision; source, full receipt and independent verification	diagnostic
R73	mission-structural	normalized-contact repair, continuous controls and shared-connection failure; full receipt and verification	diagnostic
R74	mission-active-evidence	exact finite selective-inspection reference; frozen model, exhaustive enumerator and receipt	reference
R75	library-memory	reciprocal library core without temporal boundary; three-seed acquisition, retention, cue and continuation checks with archived sources	core
R76	library-temporal-boundary	same declared library tasks with held temporal overlap; finite context, cold-acquisition failures and archived sources	core
R77	amen-patch-features	fixed-seed recorded-audio feature prediction with and without temporal overlap; matched autoregressive controls, arrays and source snapshots	core

id	manifest label	receipt	library
R78	library-selective-retention	absent-boundary, fixed-input and calibrated selective-plasticity tests; complete outcomes, checkpoints, sources and independent endpoint verification	core
R79	scalar-continuation	periodic continuation with current, temporal, explicit-history and supplied-clock inputs; frozen protocol, full receipt and arithmetic verification	core
R80	readable-trace	six scalar trace-repair lemmas; pinned sources and compiled axiom audit	formal
R81	intent-control	brief-cue motor/consequence test, matched trace ablation, private branches, continued-state acquisition and linear-decoder diagnostics	core
R82	intent-episodic	independent episodic intentions with fresh-state training; trace ablation and supplied intention/position acquisition control	core

C Experimental details

The four lives. Each life runs one process of small dense arithmetic per seed. The arm decides at 10 Hz with the torque held five body steps, plans by a beam over recorded consequences with a clipped velocity objective, and is evaluated on two isolated copies of each checkpoint, one read-only and one adapting with exploration off. The world is a 6 by 6 map with random connectivity, four walls and three doors, sixteen categorical fields with per-field missing flags, four stores with declared keys, and a best-first search over imagined cells. Connect Four uses two record cortices with receptive fields shared across positions, negamax with alpha-beta and iterative deepening inside a budget of imagined transitions, a validator that rejects an imagined board which is not the old board plus one stone, a memory of positions its searches proved and of the columns winners played, and a life of 1,200 games against random, one-ply, snapshot and solver opponents. The artist uses a 32 by 32 canvas, six pens, an eye of two windows around the gaze and the hand, and a two-level search over stroke intentions and the torques that realise them. Acceptance is seeds 10 to 14 at full budget in every case.

Learning configuration. Unless a receipt states otherwise, supervised rows use the learning model of Theorem 4.6 with $dt = 0.5$, a cross-entropy nudge at $\beta = 0.1$, a free phase of up to 100 steps and nudged phases of 12 to 50 steps at tolerance 10^{-4} to $3 \cdot 10^{-3}$. Records use a rate of 0.2 for consequence fields and 1.0 for valued fields and 0.5 to 1 percent of cells active. Baselines are trained in the same script on the same split with Adam, selected on the same validation set, and read the test set once.

Wiring search and budget curves. The four families are streams of 8,000 decisions over 24 opaque coordinates, permuted and rescaled per world seed, with a required field whose rule names one, two, or two plus a context coordinate, or the action; the oracle mask is the labelled ceiling and the floors are 0.75, 0.80, 0.55 and 0.95. The search splits a stream by whole episodes into a fitting half, a selection quarter and an untouched quarter, scores on development seeds 11 to 13 and reads acceptance seeds 21 to 23 once. The budget curves run each arm at 5,000, 20,000, 80,000 and 320,000 decisions on ten, ten, three and two lives, and read the first world back with learning off after 5,000 decisions in a second world. The evolution world runs 300 founders to a cap of 1,500 at 6,000 or 8,000 ticks, seeds 1 and 2, with the read price at 0.0004, 0.0001 or zero per unit.

Worm contact trace. The recorded encounter uses the exported trained browser model, dish seed 7 and action-sampling seed 17, selecting the first boundary contact without conditioning on the response direction. It occurs at decision 65 of the rollout, during the third search. A touch-off control starts from the identical neural state and preserves post-contact odours. All weights and biases are frozen. The 35-step run stops when the maximum activation movement falls below 0.001; the final equation residual is 0.00406. The incoming-weight contraction condition is not established for this fitted graph, so the displayed settling is an observed response rather than a certified unique equilibrium. The source snapshot, per-neuron trace and deterministic replay checks accompany the evidence [R70].

Connectome rows. The worm, fly and mouse wirings load from their public sources with pinned hashes, signs from transmitter annotations where available. The fly row imposes a row mass of 0.85 so that Theorem 4.1 applies and the residual certifies the state. The worm and mouse rows use the library’s own neuron with a gain chosen so that the net is neither silent nor saturated.

D Larger sequence and game controllers

The hypothesis is about complex multi-cortex brains, so the paper reports the largest ones trained, with the results as they stand.

The composer. The largest is a musician of 17,855 neurons in fourteen regions with 68,570,458 directed synapses and 52,849,100 trainable parameters: an ear, a sense, a mood and an interval region projecting into melody, harmony, rhythm and timbre cortices, a form cortex, a phrase cortex, a prefrontal working memory written as a trace, a recall record keyed by the bar of a sixteen-bar cycle, a slow trace of the piece so far, and an intention region of five softmax slots, all settled in one joint equilibrium and learned by the centred contrast with no backpropagation graph [R68]. It imitates a corpus of 65,188 public-domain pieces, 46.8 million events, one event at a time. Its held-out surprise is 1.023 nats per event after 38,000 updates and 5.2 hours on one accelerator; a gated recurrent unit of 7.6 million parameters trained through time on the same streams reaches 0.708 in 269 seconds. Width did not close that gap and neither did settling depth; the settling probe puts the brain’s surprise at equilibrium 0.05 below the 32-step figure and no closer. This is the loss the hypothesis has to explain, and Section 11 names the mechanism.

A second composer is built the other way round, as one life on an emulated sound chip: 2,355 settled neurons, a world head of 4,000 cells, a phrase store, a preference store, a theme store of 16,000 cells, and voice heads reading the pitch slot, with 541 motor neurons over 28 fields and an ear of 68 features per channel [R58]. On five acceptance seeds it passes ten of thirteen gates: its continuation surprise is 1.270 nats against a gate of 1.608, it beats a procedural control by 0.371, a shuffled pairing by 0.389 and its born-frozen twin by 0.407, and it decides within 18.5 ms. It fails the two musical gates, pitch within a semitone at 0.407 and onset agreement at 0.498 against 0.85, where an echo of the demonstration’s own registers reaches 0.916 and 0.924. A theme it is asked to bring back returns at 0.432 and at 0.011 with the theme store erased. Measured as curves over a life, imitation climbs from 0.10 after 6,160 blocks to 0.53 after 50,162 and stops; four times the life changes pitch by -0.037 ± 0.301 and four times the brain by -0.029 ± 0.163 , every interval covering zero; an instruction objective takes playing under a request from 0.260 to 0.620 and is flat from the first point of every life [R65]. The settled regions of that brain, workspace, dynamics, context, recall, goal and motor, take no part in a decision: the planner reads the moment and the records.

The instrument. The chip alone, learned as one life, passes all ten of its gates on five seeds: held-out prediction gain 0.591 over persistence and 0.741 on transitions, prediction retained after the matching phase at +0.021, a median pitch error of 0.0027 cents measured on the audio, silence when nothing is asked 0.991, a born-frozen tone loss of 0.798 and a shuffled-pairing loss of 0.442, at a decision latency of 8.0 ms [R57].

The player. A brain of 38,577 neurons and 1,056,874 synapses, a retina of 16,992 neurons with an afterimage of the same size, a body, an efference copy, a route memory, a context echo, foveal and peripheral feature maps of 1,800 and 1,680 neurons, an association cortex of 512 and a motor region of nine, watches 275,960 frames of a recorded expert run of the console game 1943 and learns to press the pad: 1,500 updates on one accelerator in 332 seconds take the action loss from 1.101 to 0.867 bits per slot [R69]. Frozen and set down on eight untouched starts it scores 6,062 points and survives 601 decisions on the game it watched; on 1942, a game it never watched, practice from its own play takes it from 369 to 906 points. A records controller built for the same game, 1,413 neurons with a record cortex of 16,000 cells, is measured against a convolutional actor-critic trained by proximal policy optimisation on the same view at matched frames: it is ahead by 1,359, 2,925, 475, 2,009 and 183 points at 12,500 to 200,000 decisions, its best checkpoint scores 6,175, and both sit below a scripted sweep at 7,975 to 8,850; neither clears the first stage in 1,800 held-out episodes, and the conventional arm is 9 to 35 times cheaper per decision [R66]. A brain playing sixteen console games at once with a per-stream dopamine takes 1942, which it never watched, from 350 to 2,067 points in 2,048 episodes and gives the platform games back when continued [R27].

For the hypothesis these are the scale results as they stand: a brain of seventy million synapses settles, learns and composes, and a brain of a million synapses watches a person play and then plays; on the measure that a sequence learner is built for, the sequence learner wins, and on behaviour that needs credit across many decisions, nothing here clears a stage.

E Notation

n neurons; v_i potential; $s_i = \text{act}(v_i)$ activation; W_{ij} effective weight from j to i ; c_i stimulus with bias absorbed; dt step; ρ_W largest absolute incoming weight sum; L Lipschitz constant of act; q contraction constant; v^* equilibrium; β nudge strength; $s^\pm$ nudged equilibria; Δ contrast; η learning rate; M record table; k, q keys; y value; D code width; a active cells of a code; $k(x)$ the code of a reading, which is its key; R the expansion; e eligibility trace; δ the broadcast reward error, or the size of a stimulus change where stated; γ, λ discount and trace decay; μ the row mass of added projections; ℓ, d, T layers, width and window of the transformer in the cost model.